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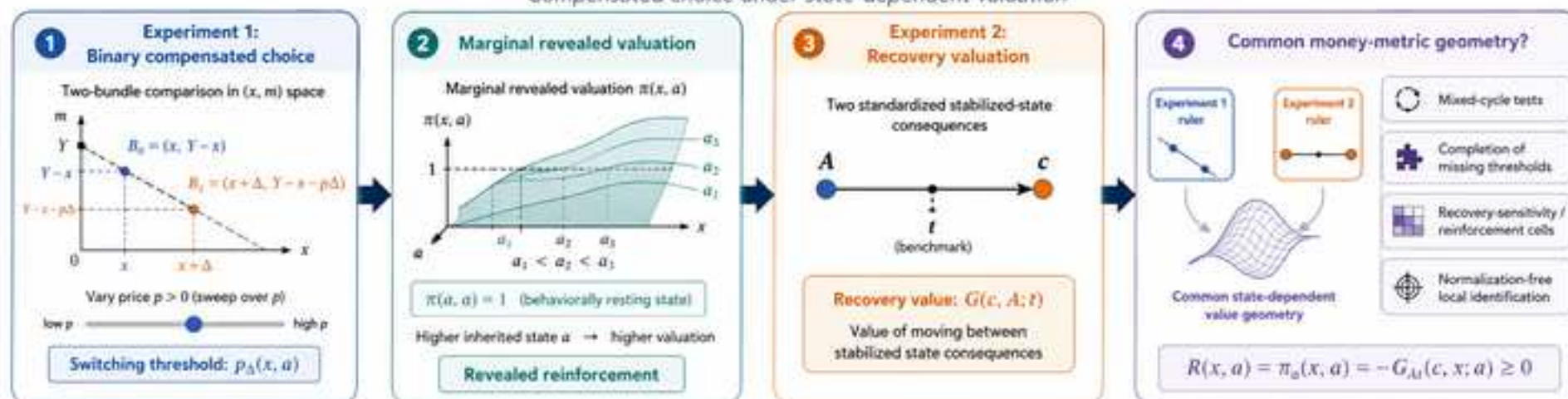
Revealed Reinforcement and Recovery Value: Compensated Choice under State-Dependent Valuation

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Abstract:	This paper develops a revealed-preference theory of state-dependent subjective valuation using compensated choice. A first experiment elicits finite switching thresholds for current consumption under inherited states; their limit defines a reservation-rate surface, with the diagonal identifying behavioral resting states and monotonicity in the inherited state defining revealed reinforcement. A second, independent experiment elicits the ordinary-good value of moving between standardized stabilized-state consequences under an assigned benchmark. The central question is whether the two experiments reveal a common money-metric valuation geometry. Finite switching chaining characterizes exact integrability of marginal thresholds, while finite recovery restrictions generate a recovery-sensitivity map and, under continuity, a positive measure. Cross-experiment compatibility then yields mixed-cycle tests, completion of missing thresholds, and normalization-free identification of state-dependence cells. In smooth environments, the same local object is identified from both experiments through the marginal state derivative and the mixed derivative of finite recovery value. The theory separates what finite choice identifies from what smooth representation adds, treating addiction as an application of a more general state-dependent valuation framework.

Revealed Reinforcement and Recovery Value

Compensated choice under state-dependent valuation



Two independent compensated-choice experiments can be cross-validated as measurements of one state-dependent subjective-value geometry.

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Professor Olivier Bochet
Editor-in-Chief
Journal of Mathematical Economics

Dear Professor Bochet,

I am submitting my manuscript, "Revealed Reinforcement and Recovery Value: Compensated Choice under State-Dependent Valuation," for consideration at the Journal of Mathematical Economics.

I would like to disclose its relationship to my previously rejected manuscript JME-D-26-00466, "Additive Measurement on Path-Generated Domains: Directed Recovery in Rational Addiction." The present paper grew out of the same broader research project, but it is not an appeal or a revision of that manuscript. The theoretical framework and main results have been substantially rebuilt.

The earlier manuscript took target-relative recovery histories as primitives, mapped them into a four-module path-generated ledger, and pursued an additive Stieltjes representation through finite cancellation, difference completeness, list Archimedean regularity, and path-generated continuity. Its central representation problem concerned scalarization on a restricted domain that was not closed under addition.

The present manuscript no longer uses that path-ledger framework or the former global extension and Archimedean route. It starts instead from two independently observed compensated-choice experiments. The first elicits finite switching thresholds for current consumption and gives a finite behavioral characterization of exact integrability through Finite Switching Chaining. The second elicits money thresholds between standardized stabilized-state consequences and characterizes their finite recovery geometry. A refutable cross-experimental compatibility condition then asks whether the two systems share one money-metric endpoint geometry, generating mixed-cycle tests, sparse completion of missing valuations, and normalization-free identification of state-dependence cells. The previous LC-A counterexample and four-module Stieltjes representation are not part of the present paper.

I am therefore submitting this as a new manuscript while disclosing the prior JME record so that the editorial team can assess the relationship directly. The manuscript is 37 pages, including appendices and references, and has been shortened in line with the journal's preference for concise submissions.

I believe the paper fits JME's emphasis on the formal mathematical expression of economic ideas: its mathematical structure is used to

answer a revealed-preference measurement question about state-dependent subjective valuation, with addiction as a leading application rather than the boundary of the theory.

The manuscript is original, has not been published, and is not under consideration elsewhere. I am the sole author and have approved the submitted version.

Thank you very much for considering the manuscript.

Yours sincerely,

Yushang Cheng

Highlights

- Binary compensation choices reveal state-conditioned reservation rates.
- A second experiment elicits values between stabilized state consequences.
- Compatibility yields mixed-cycle tests and completion of missing thresholds.
- Recovery rectangles identify state-dependence cells without utility normalization.
- Finite grids identify cell masses, not the smoothness inside cells.

Revealed Reinforcement and Recovery Value: Compensated Choice under State-Dependent Valuation

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ABSTRACT

This paper develops a revealed-preference theory of state-dependent subjective valuation using compensated choice. A first experiment elicits finite switching thresholds for current consumption under inherited states; their limit defines a reservation-rate surface, with the diagonal identifying behavioral resting states and monotonicity in the inherited state defining revealed reinforcement. A second, independent experiment elicits the ordinary-good value of moving between standardized stabilized-state consequences under an assigned benchmark.

The central question is whether the two experiments reveal a common money-metric valuation geometry. Finite switching chaining characterizes exact integrability of marginal thresholds, while finite recovery restrictions generate a recovery-sensitivity map and, under continuity, a positive measure. Cross-experiment compatibility then yields mixed-cycle tests, completion of missing thresholds, and normalization-free identification of state-dependence cells. In smooth environments, the same local object is identified from both experiments through the marginal state derivative and the mixed derivative of finite recovery value.

The theory separates what finite choice identifies from what smooth representation adds, treating addiction as an application of a more general state-dependent valuation framework.

JEL classification: D11; D12; D91; C60.

1. Introduction

Consider a long-term smoker. Holding the inherited state fixed, an investigator asks how much ordinary consumption the smoker would surrender for an additional Δ units of smoking today. Varying that ordinary-good charge until the chosen bundle switches reveals a compensation threshold. Repeating the same current-consumption comparison at different inherited states then asks a sharper question: does the same increment of smoking command a higher money value when the inherited state is higher? When it does, the inherited state has raised the subjective marginal value of current smoking. This pattern is called *revealed state reinforcement*: preference state dependence measured in ordinary-good units.

The theory begins where observation begins: with a choice between two bundles. Following the revealed-preference discipline of Samuelson (1938, 1948), and its subsequent development by Houthakker (1950); Afriat (1967); Varian (1982); Chambers and Echenique (2016), utility is asked to summarize choice rather than to precede it as evidence. The observation design, however, differs from standard demand data. Instead of observing only the bundle chosen from a changing budget, the first experiment fixes a local increase in addictive consumption and varies a numeraire transfer. Each menu therefore reveals an ordinary-good switching threshold $p_\Delta(x, a)$; letting the step shrink yields the reservation-rate surface $\pi(x, a)$. The inherited state is indexed on the same consumption-equivalent behavioral scale as current consumption, so that $\pi(a, a) = 1$ can define state a as behaviorally resting. Variation of $\pi(x, a)$ with a then measures reinforcement directly from choice.

Marginal valuation leaves open a finite question. How much does the consumer value moving between two states at which behavior has again stabilized? To answer it, the paper introduces a second, logically independent experiment. At a common elicitation date, the same decision maker chooses between standardized implementable contracts. For every benchmark t and stabilized endpoint z , a canonical consequence $K_t(z)$ fixes the timing, implementation protocol, duration, risk, transition burden, pre-implementation context, and dated ordinary-good account. Whenever the same endpoint reappears in another recovery menu, it denotes that same consequence. One contract delivers $K_t(A)$; the other

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delivers $K_t(c)$ together with an adjustable transfer in the same ordinary-good account. The transfer at which choice switches is the finite recovery value $\mathcal{G}(c, A; t)$.

The benchmark t is assigned before choice and leaves the feasible endpoints unrestricted. It therefore enters as part of the valuation protocol, much as an experimentally assigned reference condition may index a ranking without itself being chosen. The paper treats that benchmark dependence as observable structure, leaving its psychological source unrestricted. In particular, a benchmark is not inferred from the point at which movement happens to stop: doing so would mix preference with movement frictions and constraints. The state coordinate a and the benchmark t share a numerical behavioral scale, while retaining distinct economic roles.

The central economic question is whether the two experiments measure one subjective-value geometry. Exact Marginal–Finite Compatibility answers this by evaluating the marginal experiment counterfactually at state index $a = t$ and asking whether its finite money account equals the independently elicited recovery threshold under benchmark t . Equality is therefore a restriction, not a definition. When it fails, the common geometry is rejected; when it holds, marginal and recovery observations become mutually informative. Sparse observed paths then complete missing endpoint differences, while redundant paths form mixed cycles and turn identification into overidentification. The second experiment is thus valuable precisely because it is different: agreement across the two rulers has empirical content.

A second distinction concerns scale. Exact finite information need not be recoverable from a pointwise marginal limit. Moving spikes or jumps may vanish at every fixed point while leaving finite interval values unchanged. What the pointwise limit forgets, the finite account may remember. For that reason, the paper first characterizes exact finite integrability through Finite Switching Chaining and only then introduces Local Uniform Marginal Convergence to connect the finite account with the limiting calculus geometry. The recovery side is organized in the same order: finite chaining and rectangle differences identify finite spatial charges; continuity completes them into a Borel measure; smoothness, when available, yields a density and the familiar cross-partial representation.

The contribution has four connected layers. First, state-dependent marginal valuation is elicited from compensated binary choices rather than posited through a habit-stock utility function. Second, two independently observed money-value systems are tested for a common endpoint geometry, generating mixed-cycle restrictions, sparse completion, and normalization-free identification of state-dependence cells. Third, finite recovery rectangle charges are organized into a recovery-sensitivity measure, with fixed-grid data shown to identify cell masses while leaving within-cell regularity unresolved; once compatibility holds, the same cells become reinforcement cells. Fourth, local persistence, gap sufficiency, and exact target attainment are recast as further restrictions on observable compensated choices.

The framework is general state-dependent valuation, with addiction serving as the running application. Dynamic habit and addiction models explain how history generates future behavior through state transitions, intertemporal optimization, learning, self-control, or commitment (Pollak, 1970; Ryder and Heal, 1973; Becker and Murphy, 1988; Orphanides and Zervos, 1995; Gruber and Köszegi, 2001; Bernheim and Rangel, 2004; Gul and Pesendorfer, 2007). The present analysis supplies a prior measurement layer: given a state and a set of implementable consequences, what compensated choices reveal that the state changes current subjective valuation, how finite state movements are valued, and whether those two measurements speak with one voice.

The paper proceeds as follows. Section 2 relates the analysis to revealed preference, habits and addiction, and quasilinear integrability. Section 3 defines the two experiments. The subsequent sections establish finite marginal integrability, the limiting calculus representation, finite recovery characterization, cross-experiment compatibility, structural restrictions, and finite implementation. The paper then turns to economic interpretation and scope, with logical-independence examples and proofs collected in the appendices.

2. Relation to the Literature

The paper belongs to the Samuelson–Houthakker–Afriat revealed-preference tradition, where behavioral restrictions precede utility representation (Samuelson, 1938, 1948; Houthakker, 1950; Afriat, 1967; Varian, 1982; Chambers and Echenique, 2016). Its observation unit differs from standard demand data: the experimenter varies an ordinary-good transfer in a binary menu until choice switches, so a money threshold is directly observed. This connects the analysis to money-metric and quasilinear revealed preference (Fuchs-Seliger, 1989; Rochet, 1987; Brown and Calsamiglia, 2007; Makowski and Ostroy, 2013; Nocke and Schutz, 2017; Castillo and Freer, 2023). The distinctive finite question here is whether the observed thresholds themselves satisfy chaining and hence admit a common potential, rather than merely whether their marginal limit admits a calculus representation. State-specific additive constants remain free, in the spirit

of the local/global distinction emphasized by Chateauneuf and Wakker (1993); the cross-state objects used below are differences-in-differences and rectangle charges for which those constants cancel.

Habit formation and addiction usually enter through explicit dynamics linking history to current preferences (Pollak, 1970; Ryder and Heal, 1973; Becker and Murphy, 1988; Orphanides and Zervos, 1995; Gruber and Kőszegi, 2001; Bernheim and Rangel, 2004; Gul and Pesendorfer, 2007). Revealed-preference treatments of habits instead ask what finite demand observations can rationalize or identify (Crawford, 2010; Demuynck and Verriest, 2011, 2013; Crawford and Polisson, 2014). The present paper takes a prior measurement step. One experiment elicits state-conditioned compensation thresholds for current consumption; another independently elicits money thresholds between standardized stabilized-state consequences. Their agreement is tested rather than imposed. This creates cross-experimental completion, mixed-cycle overidentification, and finite state-dependence cells.

The mathematical tools have familiar relatives in quasilinear integrability, cyclic monotonicity, convex potentials, and graph-potential methods (Rochet, 1987; Krishna and Maenner, 2001; Heydenreich, Müller, Uetz and Vohra, 2009; Shiozawa, 2016). What is specific here is the economic origin of the edges and the restrictions across experiments: marginal edges come from compensated switching menus, recovery edges from independent stabilized-state thresholds, and mixed cycles test whether the two measurement systems share one endpoint-value account.

3. Environment and Revealed Choice

Choice is placed before utility representation. Let x denote current addictive consumption and let m denote ordinary consumption or saving. Normalize their relative resource price to one. The accounting identity

$$x + m = Y \quad (1)$$

describes the uncompensated baseline bundle. Experimental charges and subsidies enter separately as ordinary-good transfers, so the compensated alternative is evaluated relative to, rather than constrained by, this baseline accounting. Let x , a , and t take values in the same one-dimensional interval I . Current consumption x is measured in units of the addictive good. The inherited state a is an operational, consumption-equivalent preference-state index on that same behavioral scale. In the smoking application, a may be the previously stabilized smoking level carried into the current choice problem. Applications beginning from a primitive state s in other units can use a known behavioral index $a = \chi(s)$; the present analysis takes that index as the state coordinate and leaves the mapping χ outside the identified object. Behavioral rest is then determined from choice through the diagonal condition introduced below.

When state a is behaviorally resting, the associated bundle is

$$(x, m) = (a, Y - a). \quad (2)$$

The benchmark t is expressed on the same behavioral state scale. Predetermined and nonbinding, it leaves the endpoint set unchanged while indexing the announced valuation protocol.

Interpretation. The state variable a is an inherited preference-state coordinate affecting subjective valuation. “Reinforcement” is the revealed dependence of marginal subjective value on that state. “Recovery value” is the ordinary-good value of moving between standardized stabilized-state consequences. Stabilization on the recovery side is a property of the delivered contract; before cross-experiment compatibility is imposed, it does not itself assert the marginal resting condition $\pi(z, z) = 1$. The framework therefore separates the measurement of subjective valuation from the physiological, technological, and welfare mechanisms that may surround a particular application.

3.1. Marginal switching experiment

Fix current consumption x and inherited state a . For sufficiently small $\Delta > 0$, compare

$$B_0(x) = (x, Y - x) \quad (3)$$

with

$$B_1(x, a; p, \Delta) = (x + \Delta, Y - x - p\Delta). \quad (4)$$

The experimenter varies p . It is the local trade-off: the second bundle adds Δ units of the addictive good and removes $p\Delta$ units of the ordinary good.

Condition 3.1 (Marginal Switching Observability). *For every relevant (x, a) and all sufficiently small $\Delta > 0$, there exists a unique $p_\Delta(x, a)$ such that*

$$p < p_\Delta(x, a) \implies B_1(x, a; p, \Delta) > B_0(x),$$

$$p = p_\Delta(x, a) \implies B_1(x, a; p, \Delta) \sim B_0(x),$$

and

$$p > p_\Delta(x, a) \implies B_1(x, a; p, \Delta) < B_0(x).$$

The limit

$$\pi(x, a) := \lim_{\Delta \downarrow 0} p_\Delta(x, a) \quad (5)$$

exists and is continuous. Whenever derivative conclusions are required, the corresponding smoothness will be stated separately.

Definition 3.2 (Revealed Reservation Rate). $\pi(x, a)$ is the limiting ordinary-good reservation rate identified from the marginal switching menus. Condition 3.1 supplies the principal measurement regularity: as the ordinary-good charge varies, the binary ranking crosses indifference exactly once. That unique crossing is what cardinalizes the local comparison in ordinary-good units.

Definition 3.3 (Behaviorally Resting State). Given inherited state a , if

$$\pi(a, a) = 1, \quad (6)$$

then $(a, Y - a)$ is called a *behaviorally resting bundle*. Local choice is studied on the resulting resting locus $x = a$.

Definition 3.4 (Marginal Revealed State Reinforcement). For fixed current consumption x , the marginal data satisfy *weak revealed state reinforcement* if $a_2 > a_1$ implies

$$\pi(x, a_2) \geq \pi(x, a_1). \quad (7)$$

If every relevant comparison is strict, the data satisfy *strict revealed state reinforcement*.

If π is differentiable, weak state reinforcement implies $\pi_a \geq 0$. Strict finite ordering and a pointwise strictly positive derivative are distinct local properties.

Figure 1 summarizes the marginal measurement sequence. Binary choice first yields a finite money-metric threshold; shrinking the step then produces a state-dependent revealed-valuation surface. Each menu is evaluated on its own switching schedule, so the construction requires no additional directional restriction across separate menus.

Inherited state versus assigned benchmark. The inherited state a and the assigned benchmark t live on the same operational, consumption-equivalent behavioral scale but play different economic roles. In the marginal experiment, a is the actual inherited valuation state on which current choice is conditioned. In the recovery experiment, t indexes the announced benchmark under which stabilized-state contracts are valued. When $\pi(x, t)$ appears below, the numerical value t is inserted into the marginal state coordinate only to construct a *counterfactual marginal account at state index t* . This notation does not assert that the consumer's actual inherited state in the recovery experiment equals t . Exact Marginal-Finite Compatibility is precisely the refutable cross-context hypothesis that this counterfactual marginal account coincides with the independently observed benchmark-indexed recovery valuation.

3.2. Finite rest-to-rest experiment

Here “rest-to-rest” is protocol shorthand for comparisons between delivered stabilized consequences. The marginal condition $\pi(z, z) = 1$ is a separate behavioral statement, linked to the recovery experiment only through the joint geometry developed below.

From Binary Choice to Revealed Reinforcement

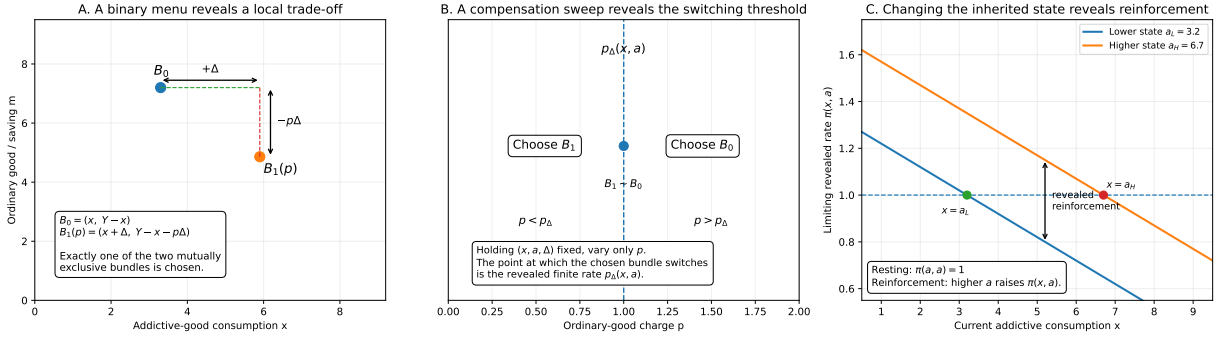


Figure 1: From binary choice to revealed reinforcement. Panel A shows the primitive marginal experiment: each elicitation compares two mutually exclusive bundles, so an increase of Δ units in addictive consumption is paid for by a reduction of $p\Delta$ units in the ordinary good. Panel B holds (x, a, Δ) fixed and varies p , identifying the finite switching rate $p_\Delta(x, a)$ from the point at which choice switches. Panel C compares the limiting rates $\pi(x, a)$ across inherited states. The diagonal crossing $\pi(a, a) = 1$ defines behavioral resting, while a higher π at the same x under a higher a reveals state reinforcement.

Canonical consequence space and choice protocol. Fix an assigned benchmark t . For every stabilized endpoint $z \geq t$, define a canonical contract consequence

$$K_t(z), \quad (8)$$

which delivers the consumer already stabilized at state z together with the baseline ordinary-good account $Y - z$. The notation suppresses ancillary contract attributes only because the protocol fixes them once and for all. In particular, the elicitation date, implementation technology, transition path or burden, duration, risk, pre-implementation inherited state, and the date and units of the ordinary-good account are held common whenever the same node $K_t(z)$ appears in any recovery comparison at benchmark t . Across benchmarks, all non-benchmark attributes are held fixed as well. Thus repeated occurrences of z denote the same economic consequence, not merely the same endpoint label.

For $t \leq c < A$, the consumer chooses at the common elicitation date between

$$A(A; t) := K_t(A) \quad (9)$$

and a transfer-adjusted lower-state contract

$$C(c, \tau; t) := K_t(c) \ominus \tau, \quad (10)$$

where $\ominus \tau$ means that τ units are subtracted from the same dated ordinary-good account used throughout the protocol. A positive τ means that the consumer is willing to surrender additional ordinary consumption, beyond the resources mechanically released by the lower addictive state, to obtain the lower stabilized consequence; a negative τ means that the lower-state contract still requires a subsidy.

This canonical-contract convention is what makes recovery-side chaining meaningful. If an intermediate endpoint b appears once as a destination and elsewhere as an origin, both appearances refer to the same consequence $K_t(b)$. Mixed marginal-recovery cycles then ask an additional question: whether marginal menus indexed by the same numerical state nodes and these canonical recovery consequences can be assigned one common endpoint-value account. That cross-context identification is supplied by E-MFC rather than by the contract convention itself. All payoff-relevant implementation attributes are therefore either standardized by the protocol or included explicitly in the consequence.

The semicolon t records the assigned valuation protocol, not an additional consumption coordinate. The recovery experiment therefore measures preference over delivered stabilized-state consequences conditional on benchmark t ; it does not infer that preference from an observed transition path. Only the ordinary-good transfer τ varies within a compensation sweep.

Condition 3.5 (Finite Rest-to-Rest Switching Observability). *For every $t \leq c < A$, there exists a unique real number $\mathcal{G}(c, A; t)$ such that*

$$C(c, \tau; t) > A(A; t) \quad \text{if } \tau < \mathcal{G}(c, A; t),$$

$$C(c, \mathcal{G}(c, A; t); t) \sim A(A; t),$$

and

$$C(c, \tau; t) < A(A; t) \quad \text{if } \tau > \mathcal{G}(c, A; t).$$

By zero-movement normalization, define $\mathcal{G}(A, A; t) := 0$. Continuity or differentiability will be imposed separately when needed.

Definition 3.6 (Revealed Finite Recovery Value). $\mathcal{G}(c, A; t)$ is the money-metric recovery value directly identified by the ordinary-good switching threshold between two delivered, already restabilized state consequences under benchmark t . It is a different experimental object from compensation for an instantaneous departure at a fixed inherited state. As in the marginal experiment, existence and uniqueness of the threshold are substantive measurement regularities that make the ranking cardinal in ordinary-good units.

4. Exact Finite-Menu Integrability

The limit π records the geometry seen as $\Delta \downarrow 0$, while the experiment also observes $p_\Delta(x, a)$ at positive finite steps. The finite data therefore ask a stronger question: do those thresholds themselves behave as finite differences of one money-metric potential? The distinction matters because a pointwise limit may be silent about value that survives on a finite interval.

For a fixed inherited state a , define the one-step *revealed money increment*

$$M_\Delta(x, a) := \Delta p_\Delta(x, a). \quad (11)$$

If $P : x = x_0 < x_1 < \dots < x_n = z$ is a partition and every step $\Delta_i = x_i - x_{i-1}$ lies in the observed finite-step domain, define its marginal account by

$$M_P(x, z; a) := \sum_{i=1}^n \Delta_i p_{\Delta_i}(x_{i-1}, a). \quad (12)$$

Definition 4.1 (Finite Switching Chaining (FSC)). For all consecutively implementable $\Delta_1, \Delta_2 > 0$ for which the three relevant step sizes are observed, require

$$(\Delta_1 + \Delta_2) p_{\Delta_1 + \Delta_2}(x, a) = \Delta_1 p_{\Delta_1}(x, a) + \Delta_2 p_{\Delta_2}(x + \Delta_1, a). \quad (13)$$

Thus the revealed money increment for moving from x to $x + \Delta_1 + \Delta_2$ in one step equals the sum of the increments obtained by splitting the same movement into two adjacent finite switching experiments.

Condition 4.2 (Local Uniform Marginal Convergence). *For every compact feasible set K , the finite switching rates converge locally uniformly to the limiting surface as $\Delta \downarrow 0$:*

$$\sup_{\substack{(x,a) \in K \\ x+\Delta \in I}} |p_\Delta(x, a) - \pi(x, a)| \longrightarrow 0. \quad (14)$$

This is the regularity that links exact finite accounts to the limiting calculus geometry.

Theorem 4.3 (Exact Finite Switching Integrability). *Fix a state a and a compact current-consumption interval J . Suppose there exists $\bar{\Delta} > 0$ such that $p_\Delta(x, a)$ is observed whenever $x, x + \Delta \in J$ and $0 < \Delta \leq \bar{\Delta}$; call a partition admissible when its mesh is at most $\bar{\Delta}$. Then the following are equivalent:*

(i) *FSC holds;*

- (ii) for every $x < z$, $M_P(x, z; a)$ is independent of the admissible partition P ;
- (iii) there exists a function $u^F(\cdot, a)$, unique up to an additive constant, such that every observed finite switching threshold satisfies

$$p_\Delta(x, a) = \frac{u^F(x + \Delta, a) - u^F(x, a)}{\Delta}. \quad (15)$$

Hence $u^F(x, a) + m$ exactly represents all observed finite binary menus at state a .

If Condition 4.2 also holds, u^F can be chosen C^1 , with

$$u_x^F(x, a) = \pi(x, a), \quad p_\Delta(x, a) = \frac{1}{\Delta} \int_x^{x+\Delta} \pi(z, a) dz. \quad (16)$$

Remark 4.4 (Same limiting surface, different finite menus). The same limiting surface π does not determine the finite switching family. Given any continuous π and $\eta \neq 0$, let

$$p_\Delta(x, a) = \frac{1}{\Delta} \int_x^{x+\Delta} \pi(z, a) dz + \eta \Delta. \quad (17)$$

Then $p_\Delta(x, a) \rightarrow \pi(x, a)$, yet the two sides of FSC differ by $2\eta\Delta_1\Delta_2$. Thus the calculus coordinate of Section 5 may exist even when the original finite- Δ menus admit no exact common potential. The first three statements of Theorem 4.3 characterize exact finite integrability; Condition 4.2 then links that finite potential to the limiting calculus geometry.

Remark 4.5 (Why finite-to-limiting regularity is separate). FSC by itself guarantees an exact finite potential, but it does not force that potential to equal the integral of π . Consider

$$u^F(x) = \mathbf{1}_{\{x \geq 0\}}, \quad p_\Delta(x) = \frac{u^F(x + \Delta) - u^F(x)}{\Delta}.$$

This family satisfies FSC exactly, and for every fixed x one has $p_\Delta(x) \rightarrow 0$, so the limiting surface is the continuous function $\pi \equiv 0$. Nevertheless, any finite interval crossing zero has a nonzero potential increment. Condition 4.2 rules out such step-dependent moving spikes and makes the exact finite account agree with the limiting integral.

5. A Calculus Representation of Marginal Choice

Even when FSC fails, the limiting marginal experiment still reveals the two-dimensional choice surface π . This section writes that limiting surface as a local calculus coordinate so that derivatives, curvature, integrals, and local expansions can be used. When FSC and Condition 4.2 both hold, Theorem 4.3 guarantees that this calculus coordinate may be chosen to coincide with the u^F that exactly represents the finite- Δ menus.

Theorem 5.1 (Calculus Representation of Revealed Switching Data). *Suppose Condition 3.1 holds and the relevant state coordinates are behaviorally resting states in the sense of Definition 3.3. Define*

$$u^R(x, a) := C(a) + \int_{\bar{x}}^x \pi(z, a) dz, \quad (18)$$

where $C(a)$ is arbitrary. Then

$$u_x^R(x, a) = \pi(x, a). \quad (19)$$

If the marginal data satisfy weak revealed state reinforcement, then:

- (i) for a fixed state a , let

$$W_a(x) := u^R(x, a) + Y - x. \quad (20)$$

Then $x = a$ is a weak maximizer of W_a ; under strict state reinforcement, it is a strict maximizer;

(ii) for every $t < x$,

$$q(x; t) := 1 - \pi(x, t) \geq 0, \quad (21)$$

and under strict reinforcement $q(x; t) > 0$.

If, in addition, $\pi \in C^1$, then

$$u_{xa}^R(x, a) = \pi_a(x, a) \geq 0, \quad q(x; t) = \int_t^x \pi_a(x, s) ds. \quad (22)$$

Thus u^R translates the resting condition and the observable state ordering into a differentiable local geometry. The objects π_a , u_{xa}^R , and the curvature results below are all expressed in this coordinate.

Corollary 5.2 (Local Diagonal Identity). *Suppose Condition 3.1 holds, the relevant diagonal states are behaviorally resting, the marginal data satisfy weak revealed state reinforcement, and $\pi \in C^1$. Then*

$$\pi_a(a, a) = -\pi_x(a, a) = u_{xa}^R(a, a) = q_x(a; a) = -q_t(a; a) \geq 0. \quad (23)$$

If, in addition, $\pi_a(a, a) > 0$, these quantities are strictly positive. Otherwise, strict finite state reinforcement may coexist with zero diagonal curvature.

6. Behavioral Characterization of Finite Recovery

This section uses only the finite thresholds $\mathcal{G}$. Finite Recovery Consistency (FRC), Finite Recovery Rectangle Positivity (FRP), and Diagonal Refinement Neutrality (DRN) respectively describe additive subdivision, positive benchmark-sensitivity rectangles, and refinement behavior near the diagonal. At this stage these are recovery-side objects only: before cross-experiment compatibility is imposed, variation with the assigned benchmark t is not yet inherited-state reinforcement. Section 7 later states the condition under which the recovery-side map is identified with the reinforcement revealed by marginal state variation.

Definition 6.1 (Finite Recovery Consistency (FRC)). Under the zero-movement normalization $\mathcal{G}(A, A; t) = 0$, finite recovery data satisfy *finite chaining* if, for all $t \leq c \leq b \leq A$,

$$\mathcal{G}(c, A; t) = \mathcal{G}(b, A; t) + \mathcal{G}(c, b; t). \quad (24)$$

Let

$$\gamma(A, t) := \mathcal{G}(t, A; t) \quad (25)$$

denote complete recovery potential. Finite chaining immediately gives

$$\mathcal{G}(c, A; t) = \gamma(A, t) - \gamma(c, t). \quad (26)$$

Thus endpoint separability is a consequence of a finite choice restriction. In particular, if $\mathcal{G}$ is C^2 , then $\mathcal{G}_{Ac} = 0$.

Definition 6.2 (Finite Recovery Rectangle Positivity (FRP)). For any $t_1 < t_2 \leq A_1 < A_2$, define

$$\begin{aligned} \Delta_{\square} \gamma &:= \gamma(A_2, t_1) - \gamma(A_1, t_1) \\ &\quad - \gamma(A_2, t_2) + \gamma(A_1, t_2). \end{aligned} \quad (27)$$

Finite Recovery Rectangle Positivity requires

$$\Delta_{\square} \gamma \geq 0 \quad (28)$$

for every admissible rectangle.

FRP is built from four finite compensation thresholds. Economically, it says that the same additional severity interval $[A_1, A_2]$ has a weakly larger revealed recovery increment under benchmark t_1 than under t_2 . This is a benchmark-sensitivity statement. It acquires the inherited-state reinforcement interpretation only if the recovery and marginal experiments satisfy the compatibility conditions of Section 7.

Definition 6.3 (Diagonal Refinement Neutrality (DRN)). For any $r < s$ and any sequence of partitions

$$P_n : \quad r = t_0^{(n)} < \dots < t_{N_n}^{(n)} = s, \quad \|P_n\| \rightarrow 0,$$

require

$$\sum_{i=1}^{N_n} \gamma(t_i^{(n)}, t_{i-1}^{(n)}) \rightarrow 0. \quad (29)$$

DRN is defined entirely from finite rest-to-rest thresholds. It excludes a purely first-order endpoint component: as a fixed interval is divided ever more finely, complete recoveries that hug the diagonal should not leave a finite first-order accumulation.

Smoothness convention. Whenever a C^1 or C^2 statement is made on the closed recovery triangle, the relevant interior derivatives are understood to extend continuously to the diagonal, equivalently as the corresponding one-sided derivatives there. No boundary derivative is used without this extension.

Theorem 6.4 (Smooth Form of the Finite Restrictions). *Suppose Finite Recovery Consistency holds and $\gamma \in C^2$. Then:*

(i) *FRP holds if and only if*

$$\gamma_{At}(A, t) = \mathcal{G}_{At}(c, A; t) \leq 0 \quad (t < A); \quad (30)$$

(ii) *DRN holds if and only if*

$$\gamma_A(t, t) = \mathcal{G}_A(t, t; t) = 0. \quad (31)$$

Moreover,

$$\Delta_{\square} \gamma = - \int_{A_1}^{A_2} \int_{t_1}^{t_2} \mathcal{G}_{At}(c, x; s) \, ds \, dx. \quad (32)$$

Theorem 6.5 (Smooth Finite-Recovery Representation). *Suppose $\mathcal{G} \in C^2$. Finite Recovery Consistency, FRP, and DRN hold if and only if there exists a unique continuous recovery-side kernel*

$$R^{\mathcal{G}}(x, a) \geq 0$$

such that

$$\mathcal{G}(c, A; t) = \int_c^A \int_t^x R^{\mathcal{G}}(x, s) \, ds \, dx. \quad (33)$$

In that case,

$$R^{\mathcal{G}}(A, t) = -\mathcal{G}_{At}(c, A; t), \quad (34)$$

and

$$q^{\mathcal{G}}(x; t) := \mathcal{G}_A(c, x; t) = \int_t^x R^{\mathcal{G}}(x, s) \, ds. \quad (35)$$

If $\mathcal{G}_{At} < 0$ whenever $t < A$, then $R^{\mathcal{G}} > 0$ off the diagonal. Continuity alone gives only $R^{\mathcal{G}}(a, a) \geq 0$; the strict inequality $R^{\mathcal{G}}(a, a) > 0$ is an additional diagonal-persistence condition.

Corollary 6.6 (What Finite Recovery Alone Reveals Locally). *Under Theorem 6.5, define for $x \geq a$*

$$\pi^{\mathcal{G}}(x, a) := 1 - q^{\mathcal{G}}(x; a). \quad (36)$$

Then $\pi^{\mathcal{G}}(a, a) = 1$, and $\pi_a^{\mathcal{G}}(x, a) = R^{\mathcal{G}}(x, a) \geq 0$ on the right-hand triangular domain. A right-sided local index constructed from $u_x^{\mathcal{G}} = \pi_x^{\mathcal{G}}$ satisfies $u_{xa}^{\mathcal{G}} \geq 0$. If $R^{\mathcal{G}}(a, a) > 0$, then $x = a$ is a strict second-order local maximum against upward deviations $x > a$. The finite recovery experiment therefore identifies the local recovery-side geometry only on the right side; two-sided conclusions require the corresponding marginal data. The constructed surface $\pi^{\mathcal{G}}$ is a recovery-side coordinate and is not identified with the marginally observed π until cross-experiment compatibility is maintained.

6.1. Finite recovery-sensitivity maps and continuous completion

FRP and DRN can be expressed directly on a finite grid. Throughout this subsection, the rectangle charges and their measure are recovery-side benchmark-sensitivity objects. The term “reinforcement” is reserved for the common object obtained after the marginal and recovery experiments pass the compatibility test. Given a partition

$$P : \quad z_0 < z_1 < \dots < z_M, \quad I_i := (z_{i-1}, z_i],$$

define, for every cell strictly below the diagonal,

$$C_{ij} := I_i \times I_j, \quad j < i,$$

its *revealed recovery rectangle charge*

$$\boxed{\rho_P(C_{ij}) := \Delta_{\square} \gamma(C_{ij}) \geq 0.} \quad (37)$$

The quantity $\rho_P(C_{ij})$ is constructed entirely from four finite recovery thresholds. It answers a more immediate economic question than the existence of a measure: *where, across severity–benchmark combinations, is recovery value concentrated?*

If P' refines P , telescoping of rectangle differences gives

$$\boxed{\rho_P(C) = \sum_{\substack{C' \subset C \\ C' \in P'}} \rho_{P'}(C').} \quad (38)$$

Finite recovery-sensitivity maps are therefore aggregation-consistent across experimental resolutions. A continuous recovery-side measure will collect these resolution-consistent cell charges into a common object.

Figure 2 also reveals why the two-dimensional recovery-side object is economically useful. Two consumers, or two environments, may have the same aggregate $\mathcal{G}(c, A; t)$ and yet place the same total recovery-side mass in very different severity–benchmark regions. Recovery data alone therefore distinguish benchmark-sensitivity geometries that an aggregate recovery value cannot distinguish. When E-MFC is maintained, those same cells receive the sharper interpretation of inherited-state reinforcement cells.

Define the open recovery triangle

$$\mathcal{T}^\circ := \{(x, s) : \underline{z} < s < x < \bar{z}\},$$

and the complete-recovery wedge

$$\Delta(t, A) := \{(x, s) : t < s < x \leq A\}.$$

Theorem 6.7 (Continuous Completion of Finite Recovery-Sensitivity Maps). *Suppose $\mathcal{G}$ is continuous and satisfies Finite Recovery Consistency, FRP, and DRN. Then there exists a unique positive, locally finite Borel measure μ on $\mathcal{T}^\circ$ such that every admissible rectangle*

$$Q = (A_1, A_2] \times (t_1, t_2], \quad t_1 < t_2 \leq A_1 < A_2,$$

From Finite Rectangles to a Recovery-Sensitivity Map

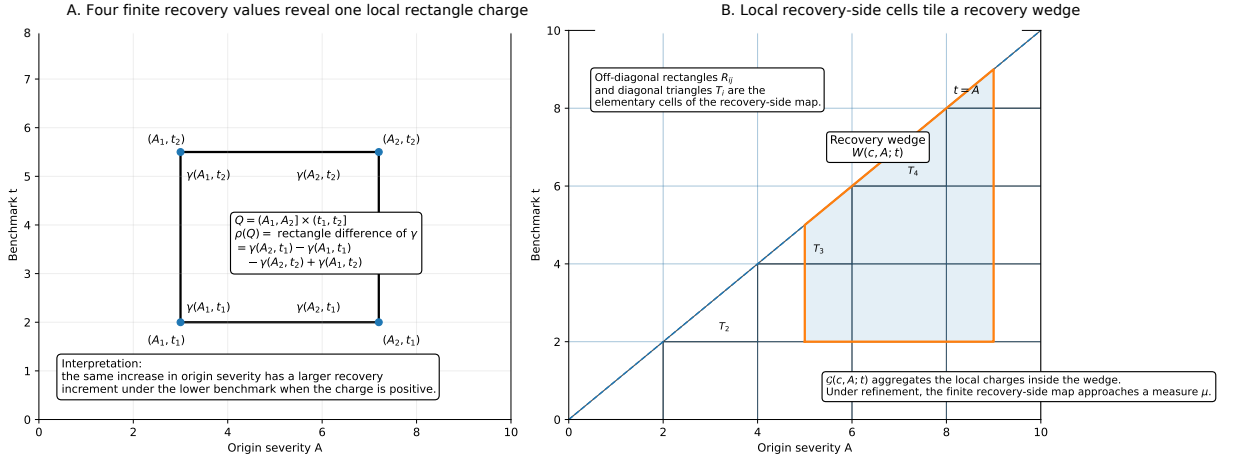


Figure 2: From finite rectangles to a recovery-sensitivity map. Panel A interprets the rectangle difference of four complete-recovery values as a local recovery-side charge, $\rho(Q) = \Delta_{\square}\gamma$. Panel B displays the two elementary cells of a finite grid: off-diagonal rectangles R_{ij} and diagonal triangles T_{ij} are the elementary cells of the recovery-side map. The value of a recovery wedge $W(c, A; t)$ is the aggregate of the local charges it contains. Under grid refinement, these resolution-consistent charges are completed in Theorem 6.7 into a positive recovery-side Borel measure μ . Before E-MFC, this measure records benchmark sensitivity of recovery valuation; under the common geometry it becomes the revealed reinforcement measure.

satisfies

$$\mu(Q) = \Delta_{\square}\gamma. \quad (39)$$

Moreover, for all interior tuples $\underline{z} < t < c \leq A < \bar{z}$,

$$\mu(\Delta(t, A)) = \gamma(A, t), \quad (40)$$

and

$$\mathcal{G}(c, A; t) = \mu(W(c, A; t)), \quad W(c, A; t) := \{(x, s) : c < x \leq A, t < s < x\}. \quad (41)$$

If, in addition, μ is absolutely continuous with respect to two-dimensional Lebesgue measure and $R := d\mu/d\lambda^2 \geq 0$, then

$$\mathcal{G}(c, A; t) = \int_c^A \int_t^x R(x, s) ds dx$$

holds with $R \in L^1_{\text{loc}}$. If further $\mathcal{G} \in C^2$, then $R = -\mathcal{G}_{At}$ almost everywhere; the continuous function $-\mathcal{G}_{At}$ is therefore a pointwise representative of the density.

Corollary 6.8 (Nested-Recovery Monotonicity). *Under Theorem 6.7, finite recovery value is nonnegative at all interior arguments and is ordered by set inclusion of recovery wedges:*

$$t < c_1 \leq c_2 \leq A \implies \mathcal{G}(c_1, A; t) \geq \mathcal{G}(c_2, A; t) \geq 0, \quad (42)$$

$$t < c \leq A_1 \leq A_2 \implies \mathcal{G}(c, A_2; t) \geq \mathcal{G}(c, A_1; t), \quad (43)$$

$$t_1 \leq t_2 < c \leq A \implies \mathcal{G}(c, A; t_1) \geq \mathcal{G}(c, A; t_2). \quad (44)$$

Each difference equals the recovery-side measure of the strip between two nested wedges. If $\mathcal{G}$ is smooth, then

$$\mathcal{G}_A \geq 0, \quad \mathcal{G}_c \leq 0, \quad \mathcal{G}_t \leq 0. \quad (45)$$

Lemma 6.9 (Triangular Telescoping Identity). *Given a partition $P : r = t_0 < t_1 < \dots < t_n = s$, let $I_i = (t_{i-1}, t_i]$. Then all grid rectangles $I_i \times I_j$ strictly below the diagonal, $j < i$, satisfy*

$$\sum_{1 \leq j < i \leq n} \Delta_{\square} \gamma(I_i \times I_j) = \gamma(s, r) - \sum_{i=1}^n \gamma(t_i, t_{i-1}). \quad (46)$$

Lemma 6.9 links the finite map to continuous completion. FRP makes every cell charge nonnegative, while DRN makes the total charge on refining grids converge to $\gamma(s, r)$ and thereby controls total recovery-side mass near the diagonal. Theorem 6.7 completes these resolution-consistent finite charges into a Borel measure.

Remark 6.10 (Why DRN matters). Rectangle increments are insensitive to an additive term $B(A) - B(c)$. DRN separates such an endpoint component from the interior recovery-sensitivity geometry through diagonal refinement.

Proposition 6.11 (Finite-Grid Regularity Nonidentification). *Fix a finite partition*

$$P : z_0 < z_1 < \dots < z_M, \quad I_i = (z_{i-1}, z_i].$$

In addition to the off-diagonal rectangles $C_{ij} = I_i \times I_j$, $j < i$, define the diagonal triangles

$$T_i := \{(x, a) : z_{i-1} < a < x \leq z_i\}.$$

Let C_P be the resulting finite cell collection, and suppose the grid-aligned recovery data identify a nonnegative mass vector $\{\rho(E) : E \in C_P\}$ whose cell sums reproduce every observed grid-aligned recovery wedge. Considered only as finite-grid data, these masses admit three positive Borel completions that agree on every cell and hence on every grid-aligned recovery threshold:

- (i) *a purely atomic completion;*
- (ii) *a singular nonatomic completion;*
- (iii) *an absolutely continuous completion with a C^∞ density.*

Thus a fixed finite recovery grid identifies spatial masses but not the within-cell regularity of an unobserved continuum completion. Continuity, absolute continuity, or smoothness of the full recovery surface are additional restrictions supplied by richer data or maintained regularity. If every elementary cell has strictly positive mass, the smooth completion can be chosen with a density that is strictly positive throughout the grid triangle.

The proposition therefore separates finite identification from continuum regularity: a grid fixes how much mass lies in each observed cell, while leaving open how that mass is distributed inside the cell.

7. Cross-Experiment Compatibility and Local Persistence

The marginal experiment supplies the entire finite switching family p_Δ and its limit π , while the finite experiment identifies $\mathcal{G}$. Cross-experiment compatibility therefore has two levels. Exact finite compatibility compares the primitive finite menus; limiting MFC compares only π and $\mathcal{G}$.

Under FSC, define for any $c < A$ the partition-independent finite marginal account

$$\mathcal{M}(c, A; a) := M_P(c, A; a), \quad (47)$$

where P is any admissible sufficiently fine partition of $[c, A]$.

Cross-experimental numeraire convention. The marginal and recovery protocols use the same dated ordinary-good unit, or a prespecified conversion into that unit. This makes the two money accounts commensurable before any compatibility restriction is tested; it does not make their valuations equal.

Economic motivation for compatibility. Both experiments assign an ordinary-good value to an endpoint difference, but they supply the conditioning state in different ways. In the marginal experiment the valuation state is inherited; in the recovery experiment a benchmark is assigned. A natural benchmark-exclusion hypothesis is that the assigned benchmark affects endpoint valuation only by selecting the same state-contingent valuation schedule that would apply counterfactually at inherited-state index $a = t$, rather than by acting as an additional taste shifter. Under that interpretation the two money accounts should agree. E-MFC tests precisely this restriction: rejection allows the benchmark to have valuation effects beyond those captured by the inherited-state geometry.

Definition 7.1 (Exact Marginal–Finite Compatibility (E-MFC)). The two finite choice data sets satisfy Exact Marginal–Finite Compatibility if FSC holds and, for all $t \leq c \leq A$,

$$\mathcal{G}(c, A; t) = (A - c) - \mathcal{M}(c, A; t). \quad (48)$$

If the one-step span $A - c$ is itself observed, this reduces to

$$\mathcal{G}(c, A; t) = (A - c)[1 - p_{A-c}(c, t)]. \quad (49)$$

E-MFC is the paper's central refutable cross-experimental hypothesis. Its left-hand side is elicited from benchmark-indexed stabilized-state contracts, while its right-hand side is constructed from marginal switching menus evaluated counterfactually at inherited-state index $a = t$. Equality says that the two independently observed money-value systems share one state-dependent endpoint-value geometry. Rejection can arise from context dependence, failure of the counterfactual state-index link, or another departure from that common geometry; because canonical recovery consequences are held fixed across menus, changing endpoint content is not among those confounds.

Theorem 7.2 (Exact Finite Cross-Experiment Representation). *Suppose FSC holds. The following are equivalent:*

- (i) *E-MFC holds;*
- (ii) *there exists a state-dependent money-metric index $u(x, a) + m$ that exactly represents all observed finite marginal menus and, under the E-MFC cross-context restriction, rationalizes the finite rest-to-rest thresholds through*

$$\mathcal{G}(c, A; t) = (A - c) - [u(A, t) - u(c, t)]. \quad (50)$$

For each state, the common index is unique up to a state-specific additive term.

When recovery sensitivity becomes reinforcement. Before compatibility is imposed, the rectangle charges and measure of Section 6 are recovery-side objects, identified from $\mathcal{G}$ alone and describing benchmark sensitivity of finite recovery valuation. Once E-MFC is maintained, the numerical benchmark coordinate t is linked to the counterfactual inherited-state coordinate of the marginal experiment. From this point onward, the common rectangle object may be called a *reinforcement cell*, and the recovery-side measure μ may be called the *revealed reinforcement measure*. The terminology therefore marks an identification delivered by compatibility.

7.1. Sparse cross-experimental completion

Theorem 7.2 describes a common representation when complete finite families are available. Actual experiments usually cover only a small set of state pairs. It is therefore useful to distinguish comparisons already identified by observed thresholds from genuinely additional observations that test the model.

Fix a benchmark t and a finite set of state nodes

$$t \leq x_1 < x_2 < \cdots < x_n, \quad \Delta_{ij} := x_j - x_i \quad (i < j).$$

For an observed marginal edge $e = (i, j, m)$, define the revealed edge increment

$$\omega_e := \Delta_{ij} p_{\Delta_{ij}}(x_i, t). \quad (51)$$

For an observed recovery edge $e = (i, j, r)$, define

$$\omega_e := \Delta_{ij} - \mathcal{G}(x_i, x_j; t). \quad (52)$$

The same pair of endpoints may carry both types of observation, so the data form a finite multigraph $\mathcal{H}_t$. Each edge is oriented from x_i to x_j ; traversing it backward in a walk reverses its sign. For a walk P , define the signed account

$$\Omega_t(P) := \sum_{e \in P} \sigma_e \omega_e, \quad \sigma_e \in \{1, -1\}. \quad (53)$$

Connectivity refers throughout to the underlying undirected multigraph.

Theorem 7.3 (Sparse Joint Rationalizability and Completion). *Fix t and the finite observation network above. The following are equivalent:*

- (i) *there exists a node potential $U_t : \{x_1, \dots, x_n\} \rightarrow \mathbb{R}$ such that every observed edge from x_i to x_j satisfies*

$$\omega_e = U_t(x_j) - U_t(x_i). \quad (54)$$

Hence the node account U_t reproduces all observed marginal and recovery switching thresholds at the observed nodes;

- (ii) *every closed walk C satisfies*

$$\Omega_t(C) = 0; \quad (55)$$

- (iii) *any two observed paths P, Q joining the same endpoints satisfy*

$$\Omega_t(P) = \Omega_t(Q). \quad (56)$$

Within each connected component, U_t is unique up to an additive constant. If x_i and x_j lie in the same component, then

$$D_t(i, j) := \Omega_t(P_{ij}) = U_t(x_j) - U_t(x_i) \quad (57)$$

is independent of the path P_{ij} . Therefore, whenever the corresponding binary menu is experimentally feasible, an unobserved marginal threshold and an unobserved recovery threshold are uniquely predicted by

$$p_{\Delta_{ij}}^{\text{pred}}(x_i, t) = \frac{D_t(i, j)}{\Delta_{ij}}, \quad \mathcal{G}^{\text{pred}}(x_i, x_j; t) = \Delta_{ij} - D_t(i, j). \quad (58)$$

Theorem 7.3 identifies relative node values separately at each benchmark t . The additive constants across benchmarks remain free, exactly as the state-specific normalization in Theorem 7.2 requires.

Corollary 7.4 (Elementary and Mixed Experimental Cycles). *Theorem 7.3 contains three elementary cases:*

- (i) *zero circulation on three marginal edges is precisely FSC;*
- (ii) *zero circulation on three recovery edges is precisely FRC;*
- (iii) *a two-edge cycle consisting of a marginal edge and a recovery edge on the same endpoints gives one-step E-MFC:*

$$\mathcal{G}(x_i, x_j; t) = \Delta_{ij} [1 - p_{\Delta_{ij}}(x_i, t)].$$

Moreover, the two experiments may generate a mixed-cycle restriction even when neither experiment forms a cycle on its own. For example, for $x_1 < x_2 < x_3$, if marginal edges (1, 2) and (1, 3) and a recovery edge (2, 3) are observed, the common geometry requires

$$\Delta_{12} p_{\Delta_{12}}(x_1, t) + \Delta_{23} - \mathcal{G}(x_2, x_3; t) = \Delta_{13} p_{\Delta_{13}}(x_1, t). \quad (59)$$

Corollary 7.5 (Minimal Exact Design). *Under the equivalent conditions of Theorem 7.3, if $\mathcal{H}_t$ has n nodes, then all node differences are point identified on the observed node set if and only if the underlying graph is connected. Any spanning tree of $n - 1$ exact edge observations identifies every $U_t(x_j) - U_t(x_i)$; observations outside the tree create overidentifying cycle restrictions. If a connected graph has m edges, the dimension of its independent cycle restrictions is $m - n + 1$.*

If two nodes lie in different connected components, their potential difference is not identified by the observed thresholds in that network: the potential on either component can be shifted independently without changing any observed edge increment.

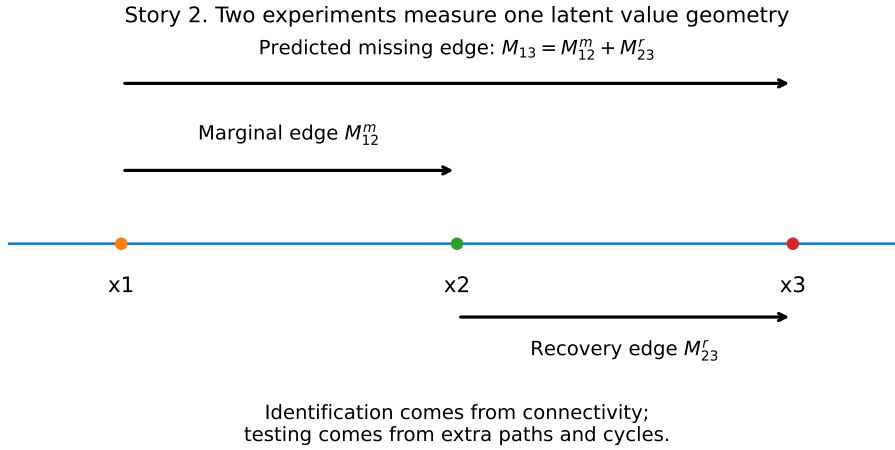


Figure 3: Two experiments, one revealed value geometry. Under joint rationalizability, marginal and recovery edges are two observations of the same endpoint potential difference in ordinary-good units. Once the mixed network is connected, observed paths complete unobserved thresholds; an additional path between already connected endpoints becomes a cycle-based overidentifying restriction. The three-node picture is the minimal intuition behind Theorem 7.3 and Corollary 7.5.

Theorem 7.6 (Normalization-Free Finite Identification of Reinforcement). *Fix an admissible rectangle*

$$Q = (A_1, A_2] \times (t_1, t_2], \quad t_1 < t_2 \leq A_1 < A_2.$$

Suppose the joint observation networks at benchmarks t_1 and t_2 satisfy Theorem 7.3, and that A_1 and A_2 are connected in each network. Define the endpoint differences identified by any compatible observed paths as

$$D_{t_j}(A_1, A_2) := U_{t_j}(A_2) - U_{t_j}(A_1), \quad j = 1, 2.$$

If the corresponding recovery data satisfy E-MFC with the common geometry, then

$$\Delta_{\square} \gamma(Q) = D_{t_2}(A_1, A_2) - D_{t_1}(A_1, A_2). \quad (60)$$

If the finite recovery data also satisfy Theorem 6.7, then

$$\mu(Q) = D_{t_2}(A_1, A_2) - D_{t_1}(A_1, A_2) = \Delta_{\square} \gamma(Q). \quad (61)$$

The difference is invariant to the separate additive normalizations of U_{t_1} and U_{t_2} . A reinforcement cell can therefore be identified by two entirely finite routes: the recovery route uses four complete-recovery thresholds; the marginal or joint-path route uses revealed endpoint differences for the same severity movement under two benchmarks. The latter route may use sparse mixed paths and need not directly observe the marginal threshold from A_1 to A_2 .

Corollary 7.7 (Smooth Limit of Finite Reinforcement Cells). *Under the common geometry of Theorem 7.6, suppose the common index u is C^2 in a neighborhood of an off-diagonal point (x, a) with $a < x$, and write $R(x, a) := u_{xa}(x, a)$. Then, as an admissible rectangle Q shrinks to (x, a) ,*

$$\frac{D_{a_2}(x_1, x_2) - D_{a_1}(x_1, x_2)}{(x_2 - x_1)(a_2 - a_1)} \longrightarrow u_{xa}(x, a) = R(x, a). \quad (62)$$

Thus the smooth cross-partial is the local limit of a finite difference-in-differences.

Definition 7.8 (Limiting Marginal–Finite Compatibility (MFC)). The limiting marginal data and finite recovery data satisfy Marginal–Finite Compatibility if, for all $t \leq c \leq A$,

$$\mathcal{G}(c, A; t) = \int_c^A [1 - \pi(x, t)] \, dx. \quad (63)$$

Corollary 7.9 (Finite and Limiting Compatibility). *Under Condition 3.1, FSC, and Condition 4.2, E-MFC holds if and only if MFC holds. The common index may then be chosen as the u^F of Theorem 4.3. If $\pi \in C^1$ and $\mathcal{G} \in C^2$, then*

$$\mathcal{G}_A(c, A; t) = 1 - \pi(A, t), \quad -\mathcal{G}_{At}(c, A; t) = \pi_a(A, t). \quad (64)$$

MFC characterizes the common calculus geometry of the limiting surface π and $\mathcal{G}$. FSC lifts that relation back to the primitive finite- Δ menus. Theorem 7.3 allows the finite observation network to be incomplete and uses mixed paths to complete missing thresholds jointly.

Theorem 7.10 (Cross-Scale Equivalence of Revealed Restrictions). *Suppose Condition 3.1, Condition 4.2, and E-MFC hold, and let*

$$\gamma(A, t) := \mathcal{G}(t, A; t).$$

Then:

(i) **Finite Recovery Consistency is automatic.** For all $t \leq c \leq b \leq A$,

$$\mathcal{G}(c, A; t) = \mathcal{G}(b, A; t) + \mathcal{G}(c, b; t). \quad (65)$$

(ii) **Resting and diagonal refinement neutrality are equivalent.** For any connected state interval J ,

$$\pi(z, z) = 1 \quad \forall z \in J \quad \Longleftrightarrow \quad \text{DRN holds on every } [r, s] \subset J. \quad (66)$$

Moreover, for any sufficiently fine partition $P : r = t_0 < \dots < t_n = s$, with $h_i = t_i - t_{i-1}$,

$$\sum_{i=1}^n \gamma(t_i, t_{i-1}) = \sum_{i=1}^n h_i [1 - p_{h_i}(t_{i-1}, t_{i-1})]. \quad (67)$$

(iii) **Finite state reinforcement and recovery-rectangle positivity are the same finite restriction.** For any admissible rectangle $t_1 < t_2 \leq A_1 < A_2$, let $P : A_1 = x_0 < \dots < x_n = A_2$ be any admissible fine partition, with $\Delta_i = x_i - x_{i-1}$. Then

$$\Delta_{\square} \gamma = \sum_{i=1}^n \Delta_i [p_{\Delta_i}(x_{i-1}, t_2) - p_{\Delta_i}(x_{i-1}, t_1)]. \quad (68)$$

If the one-step width $h = A_2 - A_1$ is directly observed, then

$$\Delta_{\square} \gamma = h [p_h(A_1, t_2) - p_h(A_1, t_1)]. \quad (69)$$

Consequently, FRP holds on all admissible rectangles if and only if finite switching rates are weakly increasing in the inherited-state coordinate throughout the recovery triangle:

$$a_2 > a_1, a_2 \leq x \implies p_\Delta(x, a_2) \geq p_\Delta(x, a_1) \quad (70)$$

for every sufficiently small observed Δ . By Theorem 4.3, this finite-step ordering holds if and only if the limiting surface satisfies $\pi(x, a_2) \geq \pi(x, a_1)$.

The full marginal experiment also contains information on the side $x < a$, which the finite recovery experiment does not cover. Strict finite switching-rate monotonicity on the recovery triangle makes every corresponding nondegenerate rectangle increment strictly positive, without implying that a smooth density must be strictly positive pointwise.

Corollary 7.11 (Joint Revealed Geometry). *Under Condition 3.1, E-MFC, and Condition 4.2, suppose the relevant state coordinates are behaviorally resting and the marginal data satisfy weak revealed state reinforcement. Then the finite recovery surface $\mathcal{G}$ is continuous and automatically satisfies FRC, FRP, and DRN. Hence:*

- (i) *in a continuous but nonsmooth environment, Theorem 6.7 yields a finite recovery-sensitivity map whose positive Borel completion is, under E-MFC, the revealed reinforcement measure compatible with the marginal restrictions;*
- (ii) *if, in addition, $\pi \in C^1$, then*

$$\mathcal{G}(c, A; t) = \int_c^A \int_t^x \pi_a(x, s) \, ds \, dx, \quad (71)$$

and at the corresponding smooth points

$$R(x, a) = \pi_a(x, a) = -\mathcal{G}_{At}(c, x; a) \geq 0. \quad (72)$$

Thus FRC, FRP, and DRN can each be tested on the finite experiment alone; under a common exact finite geometry they are also, respectively, the expressions of path decomposition, finite state reinforcement, and resting behavior.

Proposition 7.12 (Marginal Disintegration of the Reinforcement Measure). *Under the continuous common geometry of Corollary 7.11, for each current level x the function $a \mapsto \pi(x, a)$ is continuous and weakly increasing on the relevant state domain and therefore generates a unique positive Stieltjes measure ν_x ; the reinforcement measure uses its restriction to the triangular region $a < x$. The measure satisfies*

$$\nu_x((a_1, a_2]) = \pi(x, a_2) - \pi(x, a_1). \quad (73)$$

The reinforcement measure of Theorem 6.7 is uniquely characterized by

$$\mu(B) = \int \nu_x(B_x) \, dx, \quad B_x := \{a : (x, a) \in B\}, \quad (74)$$

for every Borel set $B \subset \mathcal{T}^\circ$. Equivalently, on rectangles,

$$\mu((A_1, A_2] \times (a_1, a_2]) = \int_{A_1}^{A_2} [\pi(x, a_2) - \pi(x, a_1)] \, dx. \quad (75)$$

Because π is continuous in a , each ν_x is atomless, so joint continuum data rule out point atoms in the reinforcement measure; ν_x may nevertheless be singular continuous. If $\pi \in C^1$, then

$$\nu_x(da) = \pi_a(x, a) \, da, \quad \mu(dx, da) = \pi_a(x, a) \, dx \, da. \quad (76)$$

Propositions 6.11 and 7.12 jointly define an information hierarchy. A fixed finite recovery grid identifies cell masses. A complete continuous marginal surface additionally identifies the state-Stieltjes kernels ν_x . Imposing C^1 regularity then yields the continuous density $R = \pi_a$ used by the smooth theory.

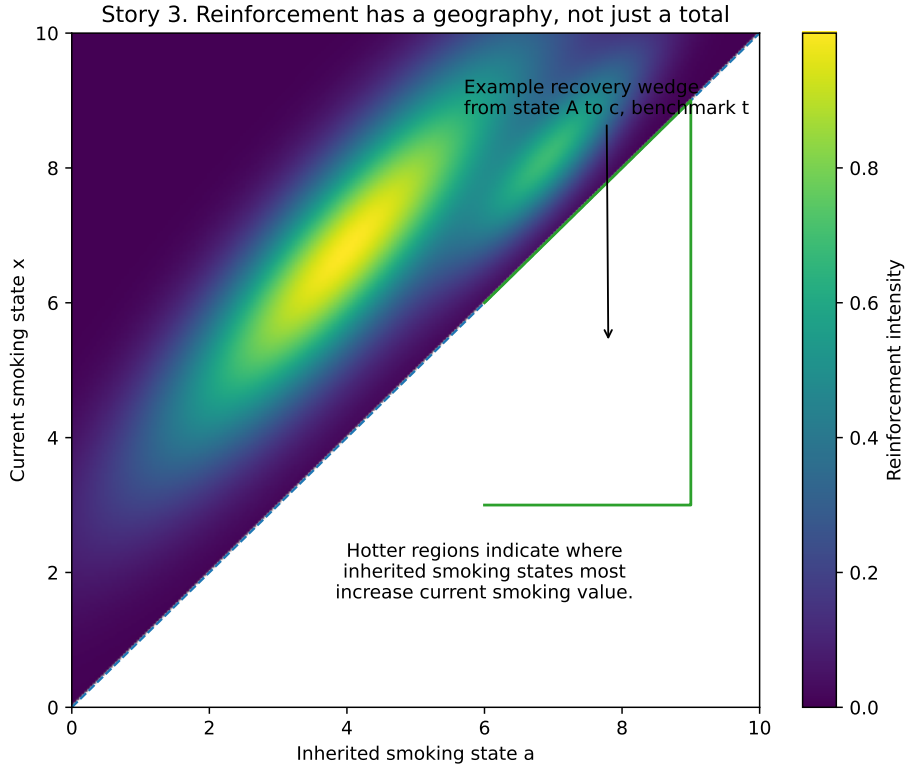


Figure 4: Illustrative smooth reinforcement geography. The heat map displays a density $R(x, a)$ under one smooth completion, making the location of reinforcement visible separately from aggregate recovery value. The figure is only an illustration of the smooth specialization. Proposition 6.11 shows that a fixed finite grid identifies only cell masses and cannot determine whether mass within a cell is atomic, singular continuous, or smooth. An ordinary density interpretation $R = \pi_a$ requires richer marginal continuum data and the corresponding regularity.

7.2. Local departure compensation

Fix inherited state a and a sufficiently small downward departure $\Delta > 0$. Compare the resting bundle

$$(a, Y - a)$$

with

$$(a - \Delta, Y - a + \Delta + D),$$

where D is an additional ordinary-good transfer.

Condition 7.13 (Local Departure Compensation Observability). *For every relevant a and all sufficiently small $\Delta > 0$, there exists a unique $D_0(a, \Delta)$ such that the two bundles are indifferent at $D = D_0(a, \Delta)$, the departure bundle is worse for $D < D_0(a, \Delta)$, and it is better for $D > D_0(a, \Delta)$.*

Assumption 7.14 (Local Common-Index Compatibility). *Let u^R be the calculus index constructed in Theorem 5.1. The local departure-compensation menus are represented by the same money-metric index*

$$U_a^R(x, m) = u^R(x, a) + m.$$

This condition links the limiting marginal geometry to the departure-compensation experiment.

Proposition 7.15 (Local Persistence Curvature). *Suppose Condition 3.1 holds, the relevant diagonal states are behaviorally resting, the marginal data satisfy weak revealed state reinforcement, $\pi \in C^1$, Condition 7.13 holds, and Assumption 7.14 is maintained. Then*

$$\lim_{\Delta \downarrow 0} \frac{D_0(a, \Delta)}{\Delta} = 0, \quad (77)$$

and

$$\kappa(a) := \lim_{\Delta \downarrow 0} \frac{2D_0(a, \Delta)}{\Delta^2} = -\pi_x(a, a) = \pi_a(a, a) \geq 0. \quad (78)$$

If diagonal marginal strictness $\pi_a(a, a) > 0$ also holds, then $\kappa(a) > 0$.

The quantity κ measures diagonal second-order persistence. Strict finite state ordering and $\kappa > 0$ are distinct local properties.

Corollary 7.16 (Tangent Gap and Local Quadratic Recovery). *Under Proposition 7.15, if Marginal–Finite Compatibility also holds, then*

$$\gamma(A, t) = (A - t) - [u^R(A, t) - u^R(t, t)]. \quad (79)$$

Consequently, as $A \downarrow t$,

$$\gamma(A, t) = \frac{1}{2}\kappa(t)(A - t)^2 + o((A - t)^2). \quad (80)$$

The quadratic term may vanish; when it does not, it is the local second-order approximation to the revealed recovery geometry.

Corollary 7.17 (Three Local Routes to Persistence). *Under Proposition 7.15 and Corollary 7.16,*

$$\kappa(t) = \pi_a(t, t) = \lim_{h \downarrow 0} \frac{2\gamma(t + h, t)}{h^2} = \lim_{\Delta \downarrow 0} \frac{2D_0(t, \Delta)}{\Delta^2}. \quad (81)$$

The same diagonal persistence curvature is therefore cross-identified by three local routes: the marginal switching surface, shrinking complete-recovery thresholds, and local departure compensation.

8. Structural Restrictions and Testable Implications

This section studies two further structures: gap sufficiency under common state translations and the boundary value of exact target attainment.

8.1. Gap sufficiency

The analysis first characterizes common-state translations in exact finite thresholds, deriving the smooth field only as a regularity corollary. Let $\mathcal{G}^\circ$ denote the recovery core excluding any exact-target boundary jump.

Definition 8.1 (Common-State Translation Invariance). Whenever all translated arguments remain feasible, the finite recovery core satisfies

$$\mathcal{G}^\circ(c + k, A + k; t + k) = \mathcal{G}^\circ(c, A; t). \quad (82)$$

The finite marginal family satisfies common-state translation invariance on the recovery half-domain $x \geq a$ if

$$p_\Delta(x + k, a + k) = p_\Delta(x, a) \quad (83)$$

for every jointly feasible (x, a, Δ, k) with $x \geq a$. The side $x < a$ remains additional information supplied only by the marginal experiment.

Theorem 8.2 (Exact Finite State Translation and a Gap Potential). *Suppose Condition 3.1 and E-MFC hold on a connected recovery-core domain closed under the relevant common translations. Then the following are equivalent:*

- (i) $\mathcal{G}^\circ$ satisfies (82);
- (ii) finite marginal thresholds satisfy (83) on the recovery half-domain;
- (iii) there exists a one-dimensional function V such that, for $d := x - a \geq 0$,

$$p_\Delta(x, a) = \frac{V(d + \Delta) - V(d)}{\Delta}; \quad (84)$$

- (iv) there exists a one-dimensional function H such that

$$\mathcal{G}^\circ(c, A; t) = H(A - t) - H(c - t). \quad (85)$$

The two one-dimensional representations may be normalized so that

$$H(d) = d - V(d), \quad V(0) = H(0) = 0. \quad (86)$$

Thus, on the recovery half-domain covered by the common geometry, translation invariance is equivalent to finite subjective-value increments depending on the state gap rather than on the absolute state level.

If, in addition, Condition 4.2 holds, the relevant diagonal points are behaviorally resting, and the marginal data satisfy weak revealed state reinforcement, then V is concave, H is convex, $H(d) \geq 0$, and $H'_+(0) = 0$.

Corollary 8.3 (Finite Gap Comparative Statics). *In the convex- H case of Theorem 8.2, write*

$$d := c - t \geq 0, \quad h := A - c > 0.$$

Then

$$\mathcal{G}^\circ(c, A; t) = H(d + h) - H(d), \quad \bar{q}(d, h) := \frac{\mathcal{G}^\circ(c, A; t)}{h}. \quad (87)$$

The finite average recovery value $\bar{q}(d, h)$ is weakly increasing in both the starting gap d and the span h . If the one-step marginal threshold is observed, E-MFC gives

$$p_h(c, t) = 1 - \bar{q}(d, h), \quad (88)$$

so $p_h(c, t)$ is weakly decreasing in both d and h .

Moreover, the complete-recovery potential H satisfies

$$\frac{H(d)}{d} \text{ is weakly increasing on } d > 0, \quad H(d_1 + d_2) \geq H(d_1) + H(d_2). \quad (89)$$

The first restriction says that average complete-recovery value rises with the target gap. The second is reinforcement compounding, not the accounting identity FRC.

Corollary 8.4 (Smooth Gap Representation). *Under the assumptions of Theorem 8.2, suppose one (and hence all) of its equivalent translation-invariance conditions holds. If the relevant diagonal points are behaviorally resting, the marginal data satisfy weak revealed state reinforcement, and $\pi \in C^1$, let $q^\circ(x; t) := 1 - \pi(x, t)$. Then there exists $r \geq 0$ such that*

$$R(x, a) = \pi_a(x, a) = r(x - a), \quad (90)$$

and, defining

$$Q(d) := \int_0^d r(z) \, dz, \quad H(d) := \int_0^d Q(y) \, dy, \quad (91)$$

we obtain

$$q^\circ(x; t) = Q(x - t), \quad \mathcal{G}^\circ(c, A; t) = H(A - t) - H(c - t). \quad (92)$$

If the relevant derivatives exist, then

$$R_x + R_a = 0, \quad q_x^\circ + q_t^\circ = 0, \quad \mathcal{G}_c^\circ + \mathcal{G}_A^\circ + \mathcal{G}_t^\circ = 0. \quad (93)$$

Remark 8.5 (The quadratic benchmark as a finite scale law). In the convex setting of Corollary 8.3, suppose the complete-recovery potential is two-homogeneous:

$$H(\lambda d) = \lambda^2 H(d) \quad (94)$$

for every jointly feasible $\lambda, d > 0$. Then $H(d) = Kd^2$ for some $K \geq 0$. In the smooth case this is equivalent to constant gap reinforcement, $r(d) \equiv 2K$, so that

$$q^\circ(x; t) = 2K(x - t), \quad \mathcal{G}^\circ(c, A; t) = K[(A - t)^2 - (c - t)^2]. \quad (95)$$

Thus quadratic recovery has a two-homogeneity characterization stated entirely in finite recovery choices.

8.2. Exact target attainment

This subsection allows an augmented exact-attainment protocol $\hat{\mathcal{G}}$. Its ordinary-good threshold is assumed to be uniquely elicited with the same switching direction as in Condition 3.5. For $c > t$ it coincides with the interior recovery core $\mathcal{G}^\circ$; at $c = t$ the experiment may additionally reveal a boundary value that appears only when the target is hit exactly. When the one-sided limit exists, define the boundary extension of the core by

$$\mathcal{G}^\circ(t, A; t) := \lim_{c \downarrow t} \hat{\mathcal{G}}(c, A; t).$$

The E-MFC prediction from the interior common geometry can therefore be separated from the exact-target observation at the boundary. Define

$$\alpha(A, t) := \hat{\mathcal{G}}(t, A; t) - \lim_{c \downarrow t} \hat{\mathcal{G}}(c, A; t). \quad (96)$$

Condition 8.6 (Target-Attainment Dominance). *Exact target attainment weakly dominates arbitrarily close attainment from above if*

$$\hat{\mathcal{G}}(t, A; t) \geq \lim_{c \downarrow t} \hat{\mathcal{G}}(c, A; t) \quad (97)$$

for every relevant (A, t) . Strict dominance replaces the inequality by $>$.

Proposition 8.7 (Target-Attainment Premium). *Suppose $\hat{\mathcal{G}}$ satisfies finite chaining for all $t \leq c \leq b \leq A$ and the one-sided limit in (96) exists. Then $\alpha(A, t)$ is independent of the origin state A . There exists a unique function $\alpha(t)$ such that*

$$\boxed{\hat{\mathcal{G}}(c, A; t) = \mathcal{G}^\circ(c, A; t) + \alpha(t)\mathbf{1}_{\{c=t\}}.} \quad (98)$$

Under Condition 8.6, $\alpha(t) \geq 0$; under strict Target-Attainment Dominance, $\alpha(t) > 0$.

Proposition 8.8 (Boundary Residual Identification). *Suppose the interior core satisfies E-MFC and let $h > 0$ be such that the one-step marginal threshold $p_h(t, t)$ is observed. Define the exact complete-recovery threshold*

$$\hat{\gamma}(t + h, t) := \hat{\mathcal{G}}(t, t + h; t).$$

Then the target-attainment premium is identified directly from two finite binary experiments:

$$\boxed{\alpha(t) = \hat{\gamma}(t + h, t) - h[1 - p_h(t, t)].} \quad (99)$$

The right-hand side must be independent of the feasible span h . Therefore, for any $h_1, h_2 > 0$, a common interior geometry requires

$$\hat{\gamma}(t + h_1, t) - h_1[1 - p_{h_1}(t, t)] = \hat{\gamma}(t + h_2, t) - h_2[1 - p_{h_2}(t, t)]. \quad (100)$$

The augmented exact-attainment protocol is exactly represented at the boundary by the same money-metric potential as the interior if and only if $\alpha(t) = 0$.

If the local common-index conditions of Proposition 7.15 and MFC also hold, then, as $h \downarrow 0$,

$$\hat{\gamma}(t + h, t) = \alpha(t) + \frac{1}{2}\kappa(t)h^2 + o(h^2). \quad (101)$$

Thus $\alpha(t)$ is the zeroth-order boundary component of exact attainment, whereas $\kappa(t)$ controls the second-order local interior component.

If the complete recovery surface, including the boundary, also satisfies common-state translation invariance, then $\alpha(t + k) = \alpha(t)$ and hence, on a connected target domain,

$$\alpha(t) \equiv \alpha. \quad (102)$$

A fully translation-invariant model therefore requires both that smooth reinforcement depend only on the gap and that the target-attainment premium be independent of the absolute target level.

Additional Structure Around the Target

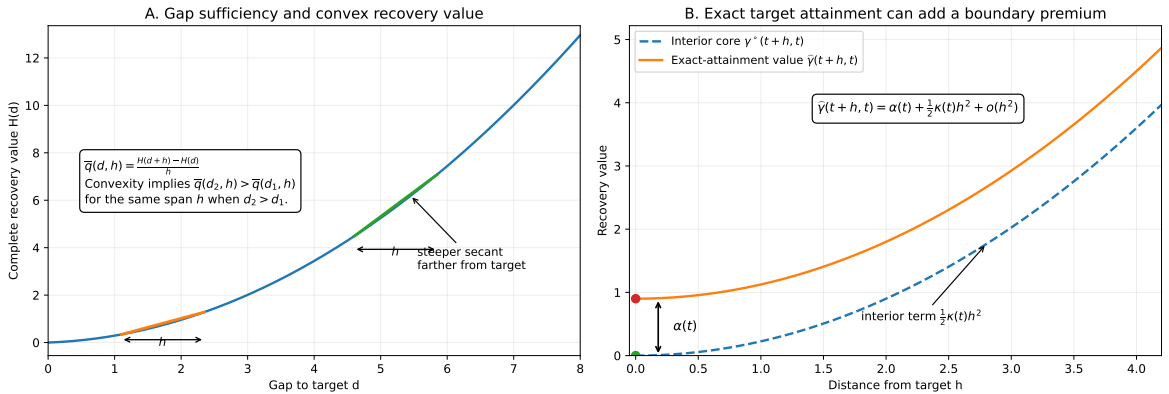


Figure 5: Additional structure around the target. Panel A shows the one-dimensional convex potential $H(d)$ under gap sufficiency: for the same recovery span h , a larger starting gap raises the average recovery value $[H(d+h) - H(d)]/h$. Panel B separates the boundary component of exact target attainment from interior persistence: $\hat{\gamma}(t+h, t) = \alpha(t) + \frac{1}{2}\kappa(t)h^2 + o(h^2)$, where $\alpha(t)$ is the zeroth-order target premium and $\kappa(t)$ controls second-order interior reinforcement.

9. Finite Implementation

9.1. Bracketed threshold completion

Theorem 7.3 uses exact thresholds. If an experiment identifies a threshold only within an interval, each observed edge $e : i \rightarrow j$ becomes a potential-difference interval

$$L_e \leq U_t(x_j) - U_t(x_i) \leq H_e. \quad (103)$$

For a marginal observation $p_{-ij} \leq p_{\Delta_{ij}}(x_i, t) \leq \bar{p}_{ij}$, set

$$L_e = \Delta_{ij}p_{-ij}, \quad H_e = \Delta_{ij}\bar{p}_{ij}; \quad (104)$$

for a recovery observation $\underline{G}_{ij} \leq \mathcal{G}(x_i, x_j; t) \leq \overline{G}_{ij}$, set

$$L_e = \Delta_{ij} - \overline{G}_{ij}, \quad H_e = \Delta_{ij} - \underline{G}_{ij}. \quad (105)$$

Replace each interval edge by two directed constraints: an edge $i \rightarrow j$ of weight H_e and an edge $j \rightarrow i$ of weight $-L_e$. Let $d_t(i, j)$ denote the shortest-path distance from i to j in the resulting constraint graph, with $+\infty$ when no directed path exists.

Proposition 9.1 (Sharp Path-Distance Completion). *The interval-threshold data are compatible with some node potential U_t if and only if the constraint graph contains no negative directed cycle. When compatible, for any x_i, x_j in the same connected component,*

$$U_t(x_j) - U_t(x_i) \in [-d_t(j, i), d_t(i, j)] \quad (106)$$

and both endpoints are sharp. Hence unobserved marginal and recovery thresholds have the sharp bounds

$$p_{\Delta_{ij}}(x_i, t) \in \left[-\frac{d_t(j, i)}{\Delta_{ij}}, \frac{d_t(i, j)}{\Delta_{ij}} \right] \quad (107)$$

and

$$\mathcal{G}(x_i, x_j; t) \in [\Delta_{ij} - d_t(i, j), \Delta_{ij} + d_t(j, i)]. \quad (108)$$

Exact observations correspond to $L_e = H_e$, in which case the path-independent point identification in Theorem 7.3 is the zero-width special case of (106).

9.2. Recovery-side map implementation

A finite recovery grid parameterizes the recovery-side rectangle measure directly by elementary cell masses. Take a state grid

$$z_0 < z_1 < \dots < z_M, \quad I_i := (z_{i-1}, z_i].$$

The elementary cells are the off-diagonal rectangles

$$C_{ij} := I_i \times I_j, \quad j < i,$$

and the triangles adjacent to the diagonal,

$$T_i := \{(x, a) : z_{i-1} < a < x \leq z_i\}.$$

Under the pure-measure representation, the data identify

$$r_{ij} := \mu(C_{ij}) = \Delta_{\square} \gamma(C_{ij}) \geq 0, \quad \delta_i := \mu(T_i) = \gamma(z_i, z_{i-1}) \geq 0. \quad (109)$$

Every grid-aligned recovery wedge is therefore a finite sum of the corresponding rectangle and diagonal-triangle masses. Collect these elementary masses as $r_1, \dots, r_J$. For target values $t_1, \dots, t_L$ appearing in the experiment, let α_ℓ denote the target-attainment premium and collect all parameters in

$$\theta = (r_1, \dots, r_J, \alpha_1, \dots, \alpha_L)'. \quad (110)$$

Every grid-aligned finite recovery design d has a known coefficient vector a_d such that

$$\hat{\mathcal{G}}(d) = a'_d \theta. \quad (111)$$

If a switching threshold is bracketed only within $[\underline{\tau}_i, \overline{\tau}_i]$, the observation contributes

$$\underline{\tau}_i \leq a'_i \theta \leq \overline{\tau}_i. \quad (112)$$

Compatible finite maps. Given finite observations $\mathcal{O}$, define

$$\Theta(\mathcal{O}) = \left\{ \theta : \begin{array}{l} r_j \geq 0, \\ \alpha_\ell \geq 0 \text{ if Condition 8.6 is maintained,} \\ \underline{\tau}_i \leq a'_i \theta \leq \bar{\tau}_i \quad \forall i, \\ H\theta = 0 \text{ for maintained linear structural restrictions} \end{array} \right\}. \quad (113)$$

If $\Theta(\mathcal{O}) = \emptyset$, the maintained structure is rejected by the finite data. If it is nonempty, it is the set of finite recovery-side maps compatible with the observations; under E-MFC these are also finite reinforcement maps.

Linear counterfactuals. For $L(\theta) = \ell'\theta$, if $\Theta(\mathcal{O})$ is nonempty and bounded, its sharp identified set is

$$\left[\min_{\theta \in \Theta(\mathcal{O})} \ell'\theta, \quad \max_{\theta \in \Theta(\mathcal{O})} \ell'\theta \right]. \quad (114)$$

The two endpoints are obtained by linear programming. For example, on a regular grid the gap-sufficiency restriction of Theorem 8.2 requires commonly translated finite thresholds and cells to obey the same gap restrictions, which appears as a set of linear equalities $H\theta = 0$. Full translation invariance additionally requires all $\alpha(t_\ell)$ to coincide.

10. Economic Interpretation and Empirical Design

This section turns the characterization into comparative and design implications. Whenever recovery-side cells or measures are interpreted as reinforcement, the common geometry—in particular E-MFC—is maintained. The identified objects can then be used to compare environments, describe heterogeneity, and allocate experimental observations.

Reinforcement dominance across environments. If two environments have reinforcement measures such that $\mu_1 - \mu_0$ is a positive measure, then for every recovery wedge,

$$\mathcal{G}_1(c, A; t) \geq \mathcal{G}_0(c, A; t).$$

Policy regimes, institutional settings, or environments can therefore be compared descriptively not only through aggregate recovery value but also through measure dominance across observable regions. A weaker local rectangle dominance orders only the corresponding class of recovery comparisons.

Same total value, different reinforcement geography. Two consumers may have the same $\mathcal{G}(c, A; t)$ on a given wedge while placing reinforcement mass in different severity–benchmark regions. Equal aggregate recovery value therefore need not mean equal state dependence. Under E-MFC, the finite recovery-sensitivity map is identified with the reinforcement map and becomes a direct object for describing this heterogeneity.

Experimental allocation of observations. Sparse completion gives a simple design principle. Spanning-tree observations identify relative subjective values; additional mixed cycles test the common geometry. If the object of interest is a particular reinforcement rectangle, Theorem 7.6 further shows that, at each benchmark, any compatible path joining the severity endpoints is sufficient to construct the marginal-route estimate of that cell, which can then be compared with the recovery-route rectangle charge.

The framework can therefore report an aggregate recovery number or decompose it into a reinforcement geography that is locatable, comparable, and overidentified by additional observations. For experiments designed to recover that geometry, the distinction between tree edges that identify and extra edges that test supplies a direct allocation principle.

11. Scope and Extensions

The theory has two observational layers. The marginal experiment reveals the finite family p_Δ and its limit π ; FSC tests whether the finite thresholds admit an exact money account. The recovery experiment reveals $\mathcal{G}$; FRC, FRP, and DRN organize those observations into an endpoint potential and a recovery-sensitivity map. E-MFC then asks whether the two layers share one finite valuation geometry. On sparse networks, mixed paths complete unobserved thresholds

and mixed cycles test the common account. Across benchmarks, endpoint differences identify reinforcement cells without fixing absolute utility levels across states.

The two experiments also have different domains of reach. Recovery comparisons cover the triangle $x \geq a$, whereas marginal menus may reveal behavior on both sides of the diagonal $x = a$. A finite recovery grid identifies cell masses; a continuous marginal surface adds the Stieltjes kernels of state variation; smoothness finally yields the density $R = \pi_a = -G_{At}$. Each refinement therefore corresponds to additional information rather than to a change in the underlying economic object.

The assigned benchmark t and inherited state a share a behavioral scale while retaining separate roles: a conditions current marginal choice, whereas t indexes the finite valuation protocol. E-MFC tests whether the counterfactual marginal account at numerical state index t matches the recovery valuation observed under benchmark t . Building on that measurement layer, one may subsequently add endogenous target formation, nonmonotone histories, repeated target crossings, or an explicit state-transition mechanism.

12. Conclusion

Compensated binary choice reveals a finite reservation threshold $p_\Delta(x, a)$ and, in the marginal limit, a state-dependent reservation-rate surface $\pi(x, a)$. Repeating the same current-consumption comparison across inherited states makes reinforcement a revealed property of choice, while $\pi(a, a) = 1$ defines behavioral rest. A separate experiment elicits the finite recovery value $G(c, A; t)$ between standardized stabilized-state consequences. Its own finite restrictions generate an endpoint potential and a recovery-sensitivity measure without presuming that benchmark sensitivity is inherited-state reinforcement.

The substantive cross-experimental hypothesis is that these two independently observed money-valued systems share one endpoint geometry. When Exact Marginal–Finite Compatibility holds, sparse paths complete missing values, redundant paths generate mixed-cycle tests, and cross-benchmark differences identify reinforcement cells without fixing state-specific utility levels. Finite observations identify increments and cell masses; smoothness adds the density $R(x, a) = \pi_a(x, a) = -G_{At}(c, x; a)$. The paper therefore supplies a measurement layer for state-dependent valuation. Dynamic state formation, endogenous targets, and welfare can be added subsequently without changing the revealed restrictions established here.

A. Logical Independence of the Stand-Alone Finite Tests

When only finite rest-to-rest data G are observed, are FRC, FRP, and DRN logically independent? The following three constructions answer yes.

Let the state interval be $I = [0, 1]$ and use the baseline

$$\gamma_0(A, t) := \frac{1}{2}(A - t)^2, \quad G_0(c, A; t) := \gamma_0(A, t) - \gamma_0(c, t). \quad (115)$$

Then G_0 satisfies FRC, every admissible rectangle increment of γ_0 equals the corresponding rectangle area, and

$$\sum_i \gamma_0(t_i, t_{i-1}) = \frac{1}{2} \sum_i (t_i - t_{i-1})^2 \leq \frac{1}{2} \|P\| (s - r) \rightarrow 0,$$

so FRP and DRN also hold.

Proposition A.1 (Independence of the Stand-Alone Finite Tests). *Under finite threshold observability and zero-movement normalization, none of FRC, FRP, and DRN is implied by the other two.*

Failure of FRC only. For any $\varepsilon > 0$, define

$$G^F(c, A; t) := G_0(c, A; t) + \varepsilon(A - c)(c - t). \quad (116)$$

The perturbation vanishes both at $c = A$ and at $c = t$, so zero recovery is preserved and

$$\gamma^F(A, t) = G^F(t, A; t) = \gamma_0(A, t).$$

Hence FRP and DRN are unchanged. But for $t \leq c < b < A$,

$$G^F(c, A; t) - G^F(b, A; t) - G^F(c, b; t) = -\varepsilon(A - b)(b - c) \neq 0, \quad (117)$$

so FRC fails.

Failure of FRP only. Let

$$\gamma^R(A, t) := (A - t)^2(1 + 4t), \quad \mathcal{G}^R(c, A; t) := \gamma^R(A, t) - \gamma^R(c, t). \quad (118)$$

FRC holds automatically by the difference form. Since $0 \leq t \leq 1$, for every partition,

$$0 \leq \sum_i \gamma^R(t_i, t_{i-1}) \leq 5\|P\|(s - r) \rightarrow 0,$$

so DRN holds. Yet for

$$t_1 = 0, \quad t_2 = \frac{1}{5}, \quad A_1 = \frac{4}{5}, \quad A_2 = 1,$$

direct calculation gives

$$\Delta_{\square} \gamma^R = -\frac{18}{125} < 0, \quad (119)$$

so FRP fails.

Failure of DRN only. Let

$$\gamma^D(A, t) := A - t, \quad \mathcal{G}^D(c, A; t) := A - c. \quad (120)$$

FRC holds and every rectangle increment is exactly zero, so FRP holds. But for every partition,

$$\sum_i \gamma^D(t_i, t_{i-1}) = s - r, \quad (121)$$

so DRN fails. □

B. Proofs

B.1. Proof of Theorem 4.3

(i) $\Rightarrow$ (ii): FSC says exactly that splitting any observed cell $[y_0, y_2]$ into two adjacent cells $[y_0, y_1]$ and $[y_1, y_2]$ leaves the account M_P unchanged. Hence any finite sequence of splittings or mergings leaves M_P unchanged. Two admissible fine partitions have a common refinement, so they generate the same account.

(ii) $\Rightarrow$ (iii): Fix a base point $\bar{x}$. For $x > \bar{x}$, define

$$u^F(x, a) - u^F(\bar{x}, a) := M_P(\bar{x}, x; a),$$

and for $x < \bar{x}$, define

$$u^F(x, a) - u^F(\bar{x}, a) := -M_P(x, \bar{x}; a).$$

Partition independence makes the definition well posed. For any observed step Δ , choose a partition containing both x and $x + \Delta$ and split at those two points. Then

$$u^F(x + \Delta, a) - u^F(x, a) = \Delta p_{\Delta}(x, a),$$

which is (15). If another function has the same finite increments, their difference is constant on the connected interval. At any menu price p , the difference between the two bundle indices is

$$[u^F(x + \Delta, a) - u^F(x, a)] - p\Delta = \Delta[p_{\Delta}(x, a) - p],$$

whose sign agrees exactly with the observed switching direction.

(iii) $\Rightarrow$ (i): Write

$$u^F(x + \Delta_1 + \Delta_2, a) - u^F(x, a)$$

first as one finite difference and then as the sum of two adjacent finite differences. This is FSC.

Finally suppose Condition 4.2 holds. Fix $x < z$ and take equally spaced partitions $x = x_0 < \dots < x_n = z$, with $h_n = (z - x)/n \leq \bar{\Delta}$. By partition independence,

$$u^F(z, a) - u^F(x, a) = \sum_{k=1}^n h_n p_{h_n}(x_{k-1}, a).$$

Local uniform convergence gives

$$\left| \sum_{k=1}^n h_n [p_{h_n}(x_{k-1}, a) - \pi(x_{k-1}, a)] \right| \leq (z - x) \sup_{y \in [x, z]} |p_{h_n}(y, a) - \pi(y, a)| \rightarrow 0.$$

Since π is continuous, the corresponding Riemann sums converge to $\int_x^z \pi(y, a) dy$. Hence

$$u^F(z, a) - u^F(x, a) = \int_x^z \pi(y, a) dy.$$

Thus u^F may be chosen C^1 with $u_x^F = \pi$. Setting $z = x + \Delta$ and using (15) yields (16). $\square$

B.2. Proof of Theorem 5.1

Choose any fixed $\bar{x}$ and any state function $C(a)$, and define

$$u^R(x, a) = C(a) + \int_{\bar{x}}^x \pi(z, a) dz.$$

Continuity of π and the fundamental theorem of calculus give $u_x^R = \pi$.

Along the resource line,

$$W_a'(x) = \pi(x, a) - 1.$$

If $x < a$, weak state reinforcement and the resting identity $\pi(x, x) = 1$ imply

$$\pi(x, a) \geq \pi(x, x) = 1,$$

so $W_a'(x) \geq 0$. If $x > a$, then

$$\pi(x, a) \leq \pi(x, x) = 1,$$

so $W_a'(x) \leq 0$. Hence $x = a$ is a weak maximizer. Under strict state reinforcement the inequalities are strict on both sides, so the maximizer is strict. Similarly, for $t < x$,

$$q(x; t) = \pi(x, x) - \pi(x, t) \geq 0,$$

with strict inequality under strict reinforcement.

If $\pi \in C^1$, differentiation under the integral gives

$$u_{xa}^R = \pi_a.$$

Weak finite monotonicity and differentiability imply $\pi_a \geq 0$, and

$$q(x; t) = \pi(x, x) - \pi(x, t) = \int_t^x \pi_a(x, s) ds.$$

$\square$

B.3. Proof of Corollary 5.2

Differentiating the identity $\pi(a, a) \equiv 1$ along the diagonal gives $\pi_x(a, a) + \pi_a(a, a) = 0$. Also $u_{xa}^R = \pi_a$ and $q = 1 - \pi$, so $q_x = -\pi_x$ and $q_t = -\pi_a$. Weak marginal state reinforcement and differentiability imply $\pi_a \geq 0$. $\square$

B.4. Proof of Theorem 6.4

Finite Recovery Consistency gives

$$\mathcal{G}(c, A; t) = \gamma(A, t) - \gamma(c, t).$$

If $\gamma \in C^2$, then $\mathcal{G}_{At}(c, A; t) = \gamma_{At}(A, t)$. The two-dimensional fundamental theorem of calculus gives

$$\Delta_{\square}\gamma = - \int_{A_1}^{A_2} \int_{t_1}^{t_2} \gamma_{At}(x, s) \, ds \, dx.$$

Thus FRP is equivalent to $\gamma_{At} \leq 0$. One direction follows immediately by integration. Conversely, if $\gamma_{At} > 0$ at some point, continuity provides a sufficiently small rectangle on which the integral has the sign that violates FRP.

For DRN, the one-dimensional fundamental theorem gives, for every partition $P : r = t_0 < \dots < t_n = s$,

$$\sum_i \gamma(t_i, t_{i-1}) = \sum_i \int_{t_{i-1}}^{t_i} \gamma_A(x, t_{i-1}) \, dx.$$

As the mesh tends to zero, uniform continuity of γ_A makes the right-hand side converge to

$$\int_r^s \gamma_A(z, z) \, dz.$$

Hence DRN holds on every $[r, s]$ if and only if the continuous function $z \mapsto \gamma_A(z, z)$ vanishes identically. Since $\mathcal{G}_A(c, A; t) = \gamma_A(A, t)$, this is (31). $\square$

B.5. Proof of Theorem 6.5

First suppose the double-integral representation holds. Splitting the outer integral at an intermediate point b gives finite chaining, and an empty integral gives zero recovery. A rectangle increment is the integral of R^G over the corresponding two-dimensional rectangle, so FRP follows. For any partition, if $R^G \leq M$ on the relevant compact interval, then

$$0 \leq \gamma(t_i, t_{i-1}) \leq \frac{M}{2}(t_i - t_{i-1})^2,$$

and therefore

$$\sum_i \gamma(t_i, t_{i-1}) \leq \frac{M}{2} \|P\| (s - r) \rightarrow 0,$$

so DRN holds.

Conversely, FRC gives $\mathcal{G} = \gamma(A, t) - \gamma(c, t)$. Theorem 6.4 yields $\gamma_{At} \leq 0$ and $\gamma_A(A, A) = 0$. Define

$$R^G(A, t) := -\gamma_{At}(A, t) = -\mathcal{G}_{At}(c, A; t) \geq 0.$$

For fixed A , diagonal neutrality gives

$$\gamma_A(A, t) = \gamma_A(A, A) - \int_t^A \gamma_{As}(A, s) \, ds = \int_t^A R^G(A, s) \, ds.$$

Integrating once more in A from c to A , and using $\mathcal{G}(c, c; t) = 0$, yields (33). Uniqueness follows from $R^G = -\mathcal{G}_{At}$. $\square$

B.6. Proof of Corollary 6.6

By Theorem 6.5, for $x \geq a$,

$$q^G(x; a) = \int_a^x R^G(x, s) \, ds,$$

so $q^G(a; a) = 0$, $\pi^G(a, a) = 1$, and $\pi_a^G = R^G \geq 0$. Along the right-hand resource line for fixed a , $W_a'(x) = \pi^G(x, a) - 1 = -q^G(x; a) \leq 0$. If $R^G(a, a) > 0$, the right-hand second derivative at a is $-R^G(a, a) < 0$. All these conclusions use only data with $x \geq a$; left-side behavior is not identified by the finite experiment. $\square$

B.7. Proof of Corollary 6.8

By Theorem 6.7, $\mathcal{C}(c, A; t) = \mu(W(c, A; t))$ and μ is a positive measure. Lowering c , raising A , or lowering t enlarges the corresponding recovery wedge by set inclusion. Equations (42)–(44) therefore follow from monotonicity of measures, and each difference is the μ -mass of the newly added strip. In the smooth case, taking the relevant one-sided derivatives of these three monotonicities gives (45). $\square$

B.8. Proof of Lemma 6.9

For fixed i , summing over $j = 1, \dots, i-1$ makes the rectangle increments telescope:

$$\sum_{j < i} \Delta_{\square} \gamma(I_i \times I_j) = \gamma(t_i, t_0) - \gamma(t_i, t_{i-1}) - \gamma(t_{i-1}, t_0),$$

where $\gamma(t_{i-1}, t_{i-1}) = 0$ has been used. Summing in turn over $i = 2, \dots, n$ telescopes the first and third terms again, leaving

$$\gamma(t_n, t_0) - \sum_{i=1}^n \gamma(t_i, t_{i-1}),$$

which is (46). $\square$

B.9. Lemmas for Theorem 6.7

Lemma B.1 (Rectangle Content Extension). *Suppose γ is continuous and satisfies FRP. On the open triangle $\mathcal{T}^\circ$, define for every admissible half-open rectangle Q*

$$v(Q) := \Delta_{\square} \gamma(Q).$$

Then there exists a unique positive, locally finite Borel measure μ such that $\mu(Q) = v(Q)$ for every admissible rectangle.

Proof. Let $\mathcal{R}$ be the semiring of admissible half-open rectangles contained in $\mathcal{T}^\circ$, and set $v(Q) = \Delta_{\square} \gamma(Q)$. Splitting a rectangle along either coordinate makes the four corner differences telescope, so v is finitely additive; FRP gives $v \geq 0$.

To obtain countable additivity on the semiring, take any decreasing sequence $Q_n \in \mathcal{R}$ with $Q_n \downarrow \emptyset$. Because all Q_n lie in the bounded first rectangle Q_1 , continuity of γ is uniform on the compact set containing their corner points. For nested half-open rectangles with empty intersection, at least one side length tends to zero. The corresponding pairs of corner evaluations of γ then coalesce, implying $v(Q_n) \rightarrow 0$. A finite additive nonnegative content on a semiring that is continuous from above at the empty set is a premeasure; hence v is countably additive on $\mathcal{R}$.

The triangle $\mathcal{T}^\circ$ has a countable cover by admissible rectangles with rational endpoints, each having finite v -mass by continuity of γ . Thus the premeasure is σ -finite. Carathéodory's extension theorem yields a unique σ -finite measure μ on $\sigma(\mathcal{R})$, the Borel σ -algebra of $\mathcal{T}^\circ$. Every compact subset of the triangle is covered by finitely many such rectangles, so μ is locally finite. $\square$

Lemma B.2 (Wedge Exhaustion from DRN). *Under Lemma B.1, if DRN holds, then for every interior pair $t < A$,*

$$\mu(\Delta(t, A)) = \gamma(A, t). \quad (122)$$

In particular, every recovery wedge has finite recovery-side mass.

Proof. Fix $t < A$ and choose a nested sequence of partitions P_n of $[t, A]$ with mesh tending to zero. Let $I_i^{(n)}$ be the cells and let U_n be the union of all $I_i^{(n)} \times I_j^{(n)}$ with $j < i$. Nested refinement gives $U_n \uparrow \Delta(t, A)$: for any $t < s < x \leq A$, the gap $x - s$ is positive, so sufficiently fine grids place s and x in different cells.

By rectangle identification and Lemma 6.9,

$$\mu(U_n) = \gamma(A, t) - \sum_i \gamma(t_i^{(n)}, t_{i-1}^{(n)}).$$

DRN makes the second term vanish. Hence $\mu(U_n) \rightarrow \gamma(A, t)$, and continuity from below gives (122). $\square$

Lemma B.3 (Recovery-Wedge Reconstruction). *If FRC holds and a positive measure μ satisfies $\mu(\Delta(t, A)) = \gamma(A, t)$, then for every $t < c \leq A$,*

$$\boxed{\mathcal{G}(c, A; t) = \mu(W(c, A; t)), \quad W(c, A; t) = \Delta(t, A) \setminus \Delta(t, c).} \quad (123)$$

Proof. FRC gives

$$\mathcal{G}(c, A; t) = \gamma(A, t) - \gamma(c, t).$$

Also $\Delta(t, c) \subset \Delta(t, A)$ and the set difference is exactly $W(c, A; t)$. Finite additivity of the measure therefore gives

$$\mathcal{G}(c, A; t) = \mu(\Delta(t, A)) - \mu(\Delta(t, c)) = \mu(W(c, A; t)).$$

□

B.10. Proof of Theorem 6.7

Lemma B.1 constructs from continuity and FRP the unique positive, locally finite rectangle-compatible measure. Lemma 6.9 supplies the discrete triangular decomposition, and Lemma B.2 uses DRN to pin the wedge mass of that rectangle measure to $\gamma(A, t)$. Finally, Lemma B.3 and FRC give (41). FRC, FRP, and DRN therefore play three distinct roles: endpoint reconstruction, positive interior mass, and diagonal mass control.

If $\mu \ll \lambda^2$, the Radon–Nikodym theorem gives $R \in L^1_{\text{loc}}$, $R \geq 0$, and Fubini’s theorem yields the double-integral representation directly. If, in addition, $\mathcal{G} \in C^2$, Theorem 6.4 together with rectangle differentiation gives $R = -\mathcal{G}_{At}$ almost everywhere. The continuous function $-\mathcal{G}_{At}$ is thus a pointwise representative of the Radon–Nikodym density. □

B.11. Proof of Proposition 6.11

Under the half-open convention, every grid-aligned recovery wedge is partitioned exactly into finitely many mutually disjoint cells $E \in C_P$. Its observed value is therefore the sum of the corresponding identified masses $\rho(E)$. Any positive measure that assigns the same mass to every elementary cell reproduces all such finite-grid observations, irrespective of how the mass is distributed within each cell.

For an atomic completion, choose a point z_E in the interior of every cell with $\rho(E) > 0$ and set

$$\mu^A := \sum_E \rho(E) \delta_{z_E}.$$

Then $\mu^A(E) = \rho(E)$.

For a singular nonatomic completion, choose in the interior of each positive-mass cell a nondegenerate smooth short curve Γ_E , let ν_E be normalized one-dimensional Hausdorff measure on that curve, and define

$$\mu^S := \sum_E \rho(E) \nu_E.$$

It has the correct mass on every cell and no atoms. Its support is a finite union of curves and therefore has two-dimensional Lebesgue measure zero, so $\mu^S \perp \lambda^2$.

For a smooth completion, choose for each positive-mass cell a function $\phi_E \in C_c^\infty(\text{int } E)$ with $\phi_E \geq 0$ and $\int_E \phi_E = 1$. Let

$$R^\infty := \sum_E \rho(E) \phi_E, \quad \mu^\infty(B) := \int_B R^\infty \, dx \, da.$$

Because the sum is finite, $R^\infty \in C^\infty$ and $\mu^\infty(E) = \rho(E)$. If every $\rho(E) > 0$, first choose a smooth strictly positive function g on the whole grid triangle and write $b_E = \int_E g > 0$. Choose $0 < \varepsilon < \min_E \rho(E)/b_E$ and define

$$R = \varepsilon g + \sum_E [\rho(E) - \varepsilon b_E] \phi_E.$$

Then $R > 0$ and the integral over every cell remains $\rho(E)$. The three completions are therefore observationally equivalent for all grid-aligned recovery data. □

B.12. Proof of Theorem 7.2

By Theorem 4.3, FSC yields an exact finite-menu potential u^F with

$$\mathcal{M}(c, A; t) = u^F(A, t) - u^F(c, t).$$

Hence E-MFC is exactly

$$\mathcal{G}(c, A; t) = (A - c) - [u^F(A, t) - u^F(c, t)],$$

which is (50). Conversely, if one common index exactly represents the finite marginal menus, each finite increment equals the corresponding difference in u , so every partition account equals $u(A, t) - u(c, t)$. The finite recovery representation then gives E-MFC. Two exact finite potentials can differ, for each state, only by an additive constant. $\square$

B.13. Proof of Theorem 7.3

(i) $\Rightarrow$ (ii): Write every edge on a closed walk as an endpoint difference of U_t . All intermediate node values telescope, so $\Omega_t(C) = 0$.

(ii) $\Rightarrow$ (iii): If P and Q join the same endpoints, then P followed by the reverse of Q is a closed walk. Hence

$$0 = \Omega_t(P \circ Q^{-1}) = \Omega_t(P) - \Omega_t(Q).$$

(iii) $\Rightarrow$ (i): Choose a root r in each connected component and set $U_t(r) = 0$. For any node x_i , choose a path P_{ri} from r to x_i and define

$$U_t(x_i) := \Omega_t(P_{ri}).$$

Path independence makes the definition well posed. For any observed edge $e : i \rightarrow j$, concatenate a path from r to i with e to obtain a path from r to j . Then

$$U_t(x_j) = U_t(x_i) + \omega_e,$$

so (54) holds. A marginal edge gives

$$U_t(x_j) - U_t(x_i) = \Delta_{ij} p_{\Delta_{ij}}(x_i, t),$$

while a recovery edge gives

$$U_t(x_j) - U_t(x_i) = \Delta_{ij} - \mathcal{G}(x_i, x_j; t).$$

Together with the switching directions in the two observability conditions, these endpoint differences reproduce the corresponding observed thresholds.

Within one component, the difference between any two compatible potentials is zero along every observed edge and therefore constant on the connected component. Equation (57) follows from path independence. Expressing the same endpoint difference in marginal and recovery units gives (58). $\square$

B.14. Proof of Corollaries 7.4 and 7.5

For $i < j < k$, zero circulation on a pure marginal triangle is

$$\Delta_{ik} p_{\Delta_{ik}}(x_i, t) = \Delta_{ij} p_{\Delta_{ij}}(x_i, t) + \Delta_{jk} p_{\Delta_{jk}}(x_j, t),$$

which is FSC. Replacing each edge value by $\Delta - \mathcal{G}$ makes the mechanical Δ terms cancel and yields FRC. A two-edge cycle formed by parallel marginal and recovery edges requires

$$\Delta_{ij} p_{\Delta_{ij}}(x_i, t) = \Delta_{ij} - \mathcal{G}(x_i, x_j; t),$$

which is one-step E-MFC. Equation (59) follows in exactly the same way from zero circulation on the corresponding mixed triangle.

If the underlying graph is connected, any spanning tree gives a unique tree path from a root to every node and therefore identifies all relative potentials. The value on any additional edge must equal the endpoint difference already determined by the tree path, so each additional edge creates a cycle restriction. A finite connected graph with m edges and n nodes has cycle-space dimension $m - n + 1$. If the graph is disconnected, adding an arbitrary constant to the potential on one component changes no observed edge difference, so potential differences across components are not identified. $\square$

B.15. Proof of Theorem 7.6 and Corollary 7.7

By Theorem 7.2, E-MFC gives a common finite potential u . For any benchmark t ,

$$\gamma(A, t) = (A - t) - [u(A, t) - u(t, t)].$$

Within a connected component of the sparse network, the potential U_t from Theorem 7.3 and $u(\cdot, t)$ reproduce the same exact edge differences. They therefore differ only by a constant, and

$$D_t(A_1, A_2) = u(A_2, t) - u(A_1, t).$$

Hence

$$\gamma(A_2, t) - \gamma(A_1, t) = (A_2 - A_1) - D_t(A_1, A_2).$$

Taking the rectangle cross-difference across t_1 and t_2 cancels the mechanical width $A_2 - A_1$ and gives (60). Under the measure representation, (39) gives $\mu(Q) = \Delta_{\square}\gamma(Q)$ and hence (61). Since each D_t uses only an endpoint difference within a fixed benchmark, adding an arbitrary constant to U_t changes nothing.

If $u \in C^2$, let $x_1 \uparrow x$, $x_2 \downarrow x$, $a_1 \uparrow a$, and $a_2 \downarrow a$. The standard limit of the average mixed difference gives

$$\frac{[u(x_2, a_2) - u(x_1, a_2)] - [u(x_2, a_1) - u(x_1, a_1)]}{(x_2 - x_1)(a_2 - a_1)} \rightarrow u_{xa}(x, a),$$

which is (62). □

B.16. Proof of Proposition 9.1

Each interval restriction (103) is equivalent to

$$U_t(x_j) \leq U_t(x_i) + H_e, \quad U_t(x_i) \leq U_t(x_j) - L_e.$$

This is a standard finite system of difference constraints. Summing along a directed cycle shows that feasibility requires every cycle length to be nonnegative. Conversely, when there is no negative cycle, shortest-path distances from an arbitrary root provide node labels satisfying all constraints.

For any compatible U_t , summing along any directed path from i to j gives

$$U_t(x_j) - U_t(x_i) \leq \text{length}(P),$$

so

$$U_t(x_j) - U_t(x_i) \leq d_t(i, j).$$

Interchanging i and j gives the lower bound $-d_t(j, i)$. To attain the upper bound, define within the component

$$U_t(x_k) := d_t(i, k).$$

The shortest-path triangle inequality guarantees all constraints, and $U_t(x_j) - U_t(x_i) = d_t(i, j)$. Taking j as the root similarly attains the lower bound. Transforming this sharp interval through $p = (U_j - U_i)/\Delta_{ij}$ and $\mathcal{G} = \Delta_{ij} - (U_j - U_i)$ yields (107)–(108). □

B.17. Proof of Corollary 7.9

By Theorem 4.3 and Condition 4.2,

$$\mathcal{M}(c, A; t) = u^F(A, t) - u^F(c, t) = \int_c^A \pi(x, t) dx.$$

Therefore E-MFC,

$$\mathcal{G}(c, A; t) = (A - c) - \mathcal{M}(c, A; t),$$

holds if and only if

$$\mathcal{G}(c, A; t) = \int_c^A [1 - \pi(x, t)] dx,$$

which is MFC. In the smooth case, differentiating with respect to A and t gives (64). □

B.18. Proof of Theorem 7.10

(i) E-MFC and partition additivity give

$$\begin{aligned}\mathcal{G}(c, A; t) &= (A - c) - \mathcal{M}(c, A; t) \\ &= (A - b) - \mathcal{M}(b, A; t) + (b - c) - \mathcal{M}(c, b; t) \\ &= \mathcal{G}(b, A; t) + \mathcal{G}(c, b; t),\end{aligned}$$

so FRC is automatic.

(ii) For a sufficiently fine partition $P : r = t_0 < \dots < t_n = s$, every cell width $h_i = t_i - t_{i-1}$ is an observed step. E-MFC gives

$$\gamma(t_i, t_{i-1}) = h_i - \mathcal{M}(t_{i-1}, t_i; t_{i-1}) = h_i[1 - p_{h_i}(t_{i-1}, t_{i-1})].$$

Summing gives (67). On the other hand, Corollary 7.9 makes E-MFC equivalent to MFC, so

$$\gamma(A, t) = \int_t^A [1 - \pi(x, t)] dx.$$

Let

$$S(P) := \sum_i \gamma(t_i, t_{i-1})$$

and compare it with the diagonal function $d(x) := 1 - \pi(x, x)$. Uniform continuity of π on the relevant compact set gives

$$\left| S(P) - \int_r^s d(x) dx \right| \leq (s - r)\omega_\pi(\|P\|) \rightarrow 0.$$

Thus DRN holds on every subinterval if and only if $d \equiv 0$, that is, if and only if $\pi(x, x) = 1$.

(iii) FRC gives

$$\Delta_{\square}\gamma = \mathcal{G}(A_1, A_2; t_1) - \mathcal{G}(A_1, A_2; t_2).$$

E-MFC then gives

$$\Delta_{\square}\gamma = \mathcal{M}(A_1, A_2; t_2) - \mathcal{M}(A_1, A_2; t_1).$$

Expanding the two accounts along any admissible partition P yields (68). If the full width h is one observed step, the single-cell partition gives (69). Hence weak monotonicity of finite switching rates in the benchmark implies FRP. Conversely, if some observed small step satisfies

$$p_{\Delta}(x, a_2) < p_{\Delta}(x, a_1), \quad a_1 < a_2 \leq x,$$

choose the rectangle $t_1 = a_1, t_2 = a_2, A_1 = x, A_2 = x + \Delta$. Equation (69) then gives a negative rectangle increment, violating FRP. The two restrictions are therefore equivalent.

Finally, by (16), if $\pi(x, a_2) \geq \pi(x, a_1)$, integrating over $[x, x + \Delta]$ immediately gives $p_{\Delta}(x, a_2) \geq p_{\Delta}(x, a_1)$. Conversely, letting $\Delta \downarrow 0$ gives weak state reinforcement of π . $\square$

B.19. Proof of Corollary 7.11

Condition 3.1 gives continuity of π , and Corollary 7.9 converts E-MFC into MFC. Hence

$$\mathcal{G}(c, A; t) = \int_c^A [1 - \pi(x, t)] dx$$

is continuous. Theorem 7.10 gives FRC, FRP, and DRN directly, so Theorem 6.7 applies. If $\pi \in C^1$, the resting identity gives

$$1 - \pi(x, t) = \pi(x, x) - \pi(x, t) = \int_t^x \pi_a(x, s) ds.$$

Substituting into MFC yields (71). The smooth conclusion of Corollary 7.9 gives

$$-\mathcal{G}_{At}(c, x; a) = \pi_a(x, a),$$

which is (72). $\square$

B.20. Proof of Proposition 7.12

For fixed x , the map $a \mapsto \pi(x, a)$ is continuous and weakly increasing on the state domain. The Lebesgue–Stieltjes construction therefore gives a unique positive measure ν_x satisfying (73). Joint continuity of π makes $x \mapsto \nu_x((a_1, a_2])$ measurable on the generating half-open intervals. A monotone-class argument then gives measurability of $x \mapsto \nu_x(B_x)$ for Borel sections, so (74) defines a positive locally finite Borel measure $\tilde{\mu}$.

For every admissible rectangle,

$$\tilde{\mu}((A_1, A_2] \times (a_1, a_2]) = \int_{A_1}^{A_2} [\pi(x, a_2) - \pi(x, a_1)] dx.$$

The finite rectangle identity in Theorem 7.10, or equivalently its limit under mesh refinement, shows that the right-hand side equals $\Delta \square \gamma$. Thus $\tilde{\mu}$ and the measure μ of Theorem 6.7 agree on a generating semiring; uniqueness implies $\tilde{\mu} = \mu$.

The Stieltjes measure of a continuous monotone function has no point atoms, so each ν_x is atomless. Equation (74) then implies that μ has no point atoms either. If $\pi \in C^1$, the Stieltjes density is $\pi_a(x, a)$, and Fubini gives (76). $\square$

B.21. Proof of Proposition 7.15

For fixed a , let $\tilde{W}_a(x) = u^R(x, a) + Y - x$. The resting identity gives $\tilde{W}'_a(a) = 0$ and

$$\tilde{W}''_a(a) = \pi_x(a, a) = -\pi_a(a, a) \leq 0.$$

Local Common-Index Compatibility gives

$$D_0(a, \Delta) = \tilde{W}_a(a) - \tilde{W}_a(a - \Delta).$$

A second-order Taylor expansion in x yields

$$D_0(a, \Delta) = -\frac{1}{2} \pi_x(a, a) \Delta^2 + o(\Delta^2),$$

which gives (77) and (78). $\square$

B.22. Proof of Corollary 7.16

By MFC,

$$\gamma(A, t) = \int_t^A [1 - \pi(x, t)] dx = (A - t) - [u^R(A, t) - u^R(t, t)].$$

The resting identity gives $u^R_x(t, t) = 1$. A second-order Taylor expansion of $u^R(A, t)$ in its first argument, together with $-u^R_{xx}(t, t) = -\pi_{xx}(t, t) = \kappa(t)$, yields (80). $\square$

B.23. Proof of Corollary 7.17

By Corollary 7.16,

$$\gamma(t + h, t) = \frac{1}{2} \kappa(t) h^2 + o(h^2),$$

so $2\gamma(t + h, t)/h^2 \rightarrow \kappa(t)$. Proposition 7.15 already gives $\kappa(t) = \pi_a(t, t)$ and the departure-compensation limit. The three routes therefore coincide. $\square$

B.24. Proof of Theorem 8.2 and Corollaries 8.3–8.4

By E-MFC, every one-step interior comparison satisfies

$$\mathcal{G}^\circ(x, x + \Delta; a) = \Delta[1 - p_\Delta(x, a)].$$

Common translation invariance of $\mathcal{G}^\circ$ therefore immediately implies common translation invariance of p_Δ . Conversely, any $\mathcal{G}^\circ(c, A; t)$ is the sum of the finite increments in an FSC account, so translation invariance of all p_Δ implies translation invariance of $\mathcal{G}^\circ$.

Suppose p_Δ is translation invariant and set $d = x - a$. By FSC, the finite account from gap 0 to gap d is independent of the partition under any benchmark a ; common translation makes it independent of the absolute value of a . Define that account as $V(d) - V(0)$ and normalize $V(0) = 0$. This gives (84). Conversely, (84) depends only on d and Δ , so the finite marginal family is translation invariant.

By E-MFC,

$$\mathcal{G}^\circ(c, A; t) = (A - c) - [V(A - t) - V(c - t)].$$

Setting $H(d) = d - V(d)$ gives (85). Conversely, if (85) holds, one-step E-MFC gives

$$p_\Delta(x, a) = 1 - \frac{H(d + \Delta) - H(d)}{\Delta} = \frac{V(d + \Delta) - V(d)}{\Delta},$$

which recovers (iii). Thus all four conditions are equivalent.

Under the additional Condition 4.2, Theorem 4.3 converts weak state reinforcement into a finite-step ordering: the fixed-step secant slopes of V are weakly decreasing in the starting gap. Hence V is concave and $H = d - V$ is convex. The resting identity gives $V'_+(0) = 1$, so $H(0) = 0$, $H'_+(0) = 0$, and convexity implies $H(d) \geq 0$.

For Corollary 8.3, the secant slope of convex H ,

$$\frac{H(d + h) - H(d)}{h},$$

is weakly increasing in both the left endpoint d and the span h , which gives the two monotonicities of $\bar{q}$. Equation (88) follows from one-step E-MFC. Convexity and $H(0) = 0$ make $H(d)/d$ weakly increasing. Monotonicity of incremental slopes also gives

$$H(d_1 + d_2) - H(d_1) \geq H(d_2) - H(0),$$

which is superadditivity.

If $\pi \in C^1$, common-state translation invariance gives $\pi(x + k, a + k) = \pi(x, a)$ and therefore $\pi(x, a) = v(x - a)$. Hence

$$R(x, a) = \pi_a(x, a) = -v'(x - a) =: r(x - a) \geq 0.$$

Resting gives $v(0) = 1$, so $1 - v(d) = \int_0^d r(z) dz = Q(d)$. Integrating once more gives $H(d) = \int_0^d Q(y) dy$, together with (92) and the PDE identities. $\square$

B.25. Proof of Proposition 8.7

Take any $t < b < A$. Finite chaining at $c = t$ gives

$$\hat{\mathcal{G}}(t, A; t) = \hat{\mathcal{G}}(b, A; t) + \hat{\mathcal{G}}(t, b; t).$$

For every $t < c < b$,

$$\hat{\mathcal{G}}(c, A; t) = \hat{\mathcal{G}}(b, A; t) + \hat{\mathcal{G}}(c, b; t).$$

Letting $c \downarrow t$ and using existence of the one-sided limit, subtracting the two equations yields $\alpha(A, t) = \alpha(b, t)$. Since b is arbitrary, the jump depends only on t . Condition 8.6 gives $\alpha(t) \geq 0$. $\square$

B.26. Proof of Proposition 8.8

The one-step E-MFC prediction of the interior core at $c = t$ is

$$\mathcal{G}^\circ(t, t + h; t) = h[1 - p_h(t, t)].$$

The boundary decomposition in Proposition 8.7 gives

$$\hat{\mathcal{G}}(t, t + h; t) = \mathcal{G}^\circ(t, t + h; t) + \alpha(t).$$

Subtracting gives (99). Proposition 8.7 has already shown that α is independent of the origin $t+h$, so the right-hand side must be the same for every feasible h , giving (100). If the augmented exact-target observation is also represented by the interior common potential, its threshold must equal the core prediction and hence $\alpha = 0$; the converse is immediate.

Finally, under the additional MFC and the conditions of Proposition 7.15, Corollary 7.16 gives the local expansion of core complete recovery,

$$\mathcal{G}^o(t, t+h; t) = \frac{1}{2}\kappa(t)h^2 + o(h^2).$$

Adding the boundary decomposition yields (101). □

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Declaration of interests

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